

The Grue Divide: When Cognitive Accuracy Doesn’t Predict Affective Alignment

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Abstract

This study presents a large-scale, exploratory investigation (9,534 trials from 1,159 participants) of color perception, cognition, and emotion in the Korean “Grue” spectrum, finding results consistent with a cognitive advantage for Korean speakers over English-speaking cohorts, particularly in Grue transitions. One illustrative case—the “Australia Divergence”—shows comparable cognitive accuracy between Korean and Australian cohorts in a preliminary, small-sample comparison (132 trials) that nonetheless appears to diverge affectively under the Sentiment-Formality-Friendliness (SFF) framework, consistent with a cognitive-affective decoupling in which perceptual precision may be shared across cultures while affective mapping remains culturally distinct. As a correlational, hypothesis-generating study, we offer these findings as a basis for future confirmatory research rather than as definitive claims, positioning SFF as a promising exploratory tool for capturing culturally mediated “emotional fingerprints” that traditional cognitive metrics may miss, though further validation against human ratings remains needed.

1 Introduction

Does our language expand our perception? The Sapir-Whorf hypothesis suggests it might, influencing even how we see the world—consider classic examples regarding the blue spectrum in Russian or unique green perceptions among Koreans. But can Large Language Models (LLMs), trained primarily on text, truly reflect these deep-seated cultural nuances, especially concerning emotion? This study explores that very question, focusing on the influence of linguistic and cultural factors on color perception and emotion within the challenging blue-green spectrum for Korean speakers, while exploring potential human behavioral markers to inform the evaluation of cultural pragmatic capabilities in LLMs.

1.1 Background and Motivation

As exemplified by the ability of Russian speakers to distinguish finer shades of blue compared to English speakers (Winawer et al., 2007), color perception is often deeply tied to linguistic frameworks. This aligns with the view that challenges biological universality, arguing that color categorization is fundamentally contingent upon social and cultural contexts (Saunders and van Brakel, 1997). This perspective is further supported by evidence demonstrating that language directly influences perception (Gilbert et al., 2006). Notably, research has found that the right visual field (RVF, processed by the language-dominant left hemisphere) plays a key role in this effect.

Recent studies have expanded the affective domain by relating various emotional terms to specific colors (Mohammad, 2013; Sutton and Altarriba, 2016). While research has begun to emphasize diversity across languages and cultures (Ananthram et al., 2023), the cultural and pragmatic nuances of Korean color terms remain underexplored. Unlike Western additive or trichromatic models, the broader Korean color lexicon is rooted in traditional frameworks such as *Obangsaek* (Five Cardinal Colors), systematically encoding multi-modal mappings across orientation, taste, sound, and affect (Kang, 2009). Within this system, the “Grue” (blue-green, *pureuda*) spectrum occupies a particularly distinct cognitive position: empirical evidence shows that while it possesses the fewest derivational variants among core color categories, it exhibits the highest free-recall frequency in spontaneous language production (Lee and Kim, 1996). This marked asymmetry between lexical economy and cognitive salience makes the Grue spectrum a critical locus for probing cross-linguistic pragmatic divergence. This study addresses this gap by applying the sociocultural framework as an exploratory tool to capture the ‘emotional fingerprints’ of Korean speakers.

1.2 Research Questions and Contributions

This study investigates the cognitive and affective dimensions of color perception across Korean and English speakers, with a primary focus on the **Grue spectrum**. By bridging objective visual discrimination tasks with computational affective modeling, we examine how linguistic frameworks shape both perceptual precision and cultural-emotional associations.

The investigation is guided by two central research questions:

1. **RQ1 (Study 1: Cognitive Perception):** How do Korean and English-speaking cohorts differ in error rate and response duration within the continuous Grue spectrum?
2. **RQ2 (Study 2: Affective Modeling):** How can the socio-linguistic and affective nuances of these color categories be captured and modeled using the proposed Sentiment-Formality-Friendliness (SFF) framework—a three-dimensional vector derived via LLM prompt-based evaluation?

This study provides an exploratory human behavioral reference point, offering preliminary data to inform future evaluations of the pragmatic gap in AI models. We explicitly frame this work as exploratory and correlational: it is designed to surface empirical patterns that merit further, more targeted confirmatory investigation, rather than to establish causal mechanisms. Our contributions are threefold: (i) **Proposal of the SFF Framework**, introducing a multi-dimensional tool to explore the socio-cultural and affective positioning of color terms; (ii) **Exploratory Evidence for Linguistic Relativity**, providing large-scale empirical data illustrating how the Korean “Grue” category is associated with cognitive and affective response patterns distinct from English-speaking cohorts; and (iii) **Synthesis of Nuanced Affective Data**, utilizing open-ended keyword responses to capture the culturally situated nuances of color perception, thereby contextualizing quantitative findings with rich behavioral data and laying the groundwork for future confirmatory validation.

2 Related Works

2.1 Linguistic Relativity and Color Perception

The relationship between language and color perception has been a central debate in cognitive linguistics. [Berlin and Kay \(1991\)](#) proposed

universal color categories, while subsequent work challenged this by demonstrating language-specific perceptual advantages. [Winawer et al. \(2007\)](#) showed that Russian speakers discriminate blues faster when stimuli cross the *siniy/goluboy* boundary, providing empirical support for the Sapir-Whorf hypothesis in color cognition. [Gilbert et al. \(2006\)](#) further demonstrated that this effect is lateralized to the right visual field, implicating language-dominant hemispheric processing.

2.2 The Korean Grue Spectrum

Korean presents a particularly compelling case for linguistic relativity through its *Grue* spectrum. [Deshpande \(2015\)](#) examined color term-perception relationships across the full color spectrum in Korean-English bilinguals, demonstrating higher sensitivity to brightness variations within the blue spectrum. However, their study focused primarily on lexical-perceptual mappings across the broader color vocabulary, rather than isolating the Grue spectrum as a distinct categorical boundary. Generational shifts in color naming further complicate this picture: younger Koreans favor native terms while older speakers retain Sino-Korean or Japanese-influenced vocabulary ([Tyson, 1998](#)). [Lee and Kim \(1996\)](#) confirmed through 42,000+ free-recall responses that the Grue spectrum remains the most actively utilized color category in Korean. For detailed cultural and linguistic background, see Appx. A.

2.3 Affective Dimensions of Color and Multicultural AI

While cognitive approaches to color perception are well-established, the affective dimensions remain underexplored in LLMs. Models trained predominantly on English-centric corpora show systematic gaps in multicultural pragmatic competence: even fluent multilingual generation fails to reflect cross-cultural differences in emotional expression ([Havaldar et al., 2023](#)), LLMs are quantitatively biased toward Western-centric value systems ([Tao et al., 2024](#)), and models implicitly treat one culture as a “standard” against which others are othered ([Li et al., 2024](#)). Translation alone is also an unreliable proxy for multicultural competence, since cultural context itself introduces noise into cross-lingual evaluation ([Singh et al., 2025](#)). Compounding this, existing sentiment analysis frameworks typically reduce emotional valence to a single dimension, missing socially-coded axes such as formality and interpersonal warmth. These gaps carry concrete

costs in multilingual deployment: translation or content generation that misses this socio-affective context—e.g., an image or passage that reads as culturally generic or contextually foreign rather than locally situated—is liable to be judged as lower-utility output in that market. Grounding this problem in a single, tightly-scoped semantic domain (color) offers a tractable starting point for identifying how generative models might be aligned to produce this kind of locally-situated output instead.

2.4 Research Gap

Prior work has largely treated cognitive and affective dimensions as separate objects of inquiry. Winawer et al. (2007) demonstrated language-specific perceptual advantages but was limited by a relatively small sample size and employed a verbal interference paradigm that restricts ecological validity and scalability. Deshpande (2015) expanded the scope to Korean-English bilinguals but did not address affective dimensions.

The present study addresses these gaps in four ways. First, we adopt a large-scale web-based platform to recruit native speakers across multiple countries, substantially increasing sample size compared to prior laboratory studies. Second, we omit the verbal interference paradigm and instead integrate a keyword-based affective elicitation task within the same experimental session. Third, by combining cognitive discrimination data with SFF-scored affective responses from the same participants, we enable direct cross-cohort comparison of both perceptual and emotional dimensions within a unified research design. Fourth, and crucially, we propose the SFF (Sentiment-Formality-Friendliness) framework as a scalable exploratory methodology for capturing culturally mediated emotional fingerprints, highlighting the potential of natively sourced affective data to inform multicultural alignment in LLMs.

3 Research Method

To facilitate dynamic data collection and a customized user experience, the experimental platform was developed using **Lovable.dev**. This approach enabled the rapid prototyping of a sophisticated survey interface and ensured seamless data integration through direct HTML customization. By leveraging this agile development framework, we created a robust experimental environment tailored to the study’s specific requirements for cross-cultural color-emotion mapping. The source code for the experimental platform (English

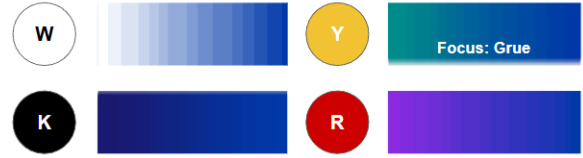


Figure 1: Representative samples of color gradient bars used in the experiment, ranging from Blue (center) to White, Black, Yellow, and Red.

deployment) is available at https://github.com/bakeyeon/colorfish_public.

3.1 Study 1: Cognitive Perception

The color perception test consists of six trials, each featuring a color gradient bar composed of 8 to 30 adjacent blocks. Centered on the blue spectrum and informed by the five cardinal colors (*Obangsaek*), these gradients represent transitions from blue to the other four cardinal colors (Fig. 1):

1. Blue to White: #0038a8 → #ffffff
2. Blue to Black: #0038a8 → #191970
3. Blue to Yellow: #0038a8 → #008b8b
4. Blue to Red: #0038a8 → #8a2be2

Stimuli were generated by linear interpolation in RGB space between two endpoint colors; more blocks meant a smaller, fixed per-step RGB distance between adjacent blocks. Precision was measured across three difficulty levels: **(1) Level 1 (Basic)**, 8–17 blocks with distinct visual boundaries; **(2) Level 2 (Subtle)**, 15–25 blocks, increased density, no boundary hints; and **(3) Level 3 (Advanced)**, 20–30 blocks, yielding the smallest per-block color steps and most ambiguous boundaries. For each trial, the block count was sampled uniformly at random from the relevant level’s range (e.g., an integer drawn from [15, 25] for Level 2). The per-block color step also depends on the specific endpoint-color pair, since a larger overall color distance yields larger per-step increments for the same block count; a Subtle trial with a high-contrast pair (e.g., blue to white) can thus in principle be easier than an Advanced trial with a lower-contrast pair. Block-count ranges for adjacent levels overlap accordingly, so that L_d in Eq. 1 reflects the intended combination of block count, boundary-hint presence, and endpoint-color contrast, rather than block count alone. To mitigate order effects, the six trials were presented in a randomized sequence for each participant using a **Fisher-Yates shuffle** ($O(n)$), with response time recorded automatically. Given the stochastic block generation, each participation instance yielded unique data points, maintaining statistical

independence even under repeated participation. To measure color perception accuracy, we calculated a normalized Estimate Error Score ($\mathbf{R}_E$) for each trial, representing the participant’s relative error rate weighted by task difficulty:

$$\mathbf{R}_E = \frac{|\mathbf{B}_a - \mathbf{E}_u|}{\mathbf{B}_a \times \mathbf{L}_d} \quad (1)$$

where $\mathbf{B}_a$ is the actual block count, $\mathbf{E}_u$ is the participant’s estimate, and $\mathbf{L}_d$ is the difficulty level (1–3). Full analysis strategy is detailed in Appx. B.2.1; a robustness check confirming our findings are insensitive to the specific choice of difficulty weights is provided in Appx. C (“ R_E Difficulty-Weighting Robustness Check”).

Grounding the Block-Counting Task in Numerosity Estimation Unlike discrimination paradigms that measure reaction time to a small, fixed set of comparison colors (Winawer et al., 2007; Gilbert et al., 2006), our task asks participants to enumerate discrete blocks along a continuous color gradient—a numerosity-estimation task. This is grounded in evidence that the visual system possesses a dedicated, non-verbal capacity for numerosity, measurable independently of contrast, size, or texture density (Burr and Ross, 2008), though such estimates remain sensitive to display cues like density and convex hull (Gebuis and Reynvoet, 2012). The design also better matches our linguistic target: Winawer et al. (2007) examined Russian, where *siniy* and *goluboy* are lexically independent, underived terms, so comparing a small set of fixed swatches suffices to probe the boundary between them. Korean’s Grue spectrum instead arises through productive morphological derivation from a single root (Appx. A.1), so we use a continuous, block-subdivided stimulus as a closer structural analogue to this derivational mechanism—a design rationale, not an empirically established mapping.

3.2 Study 2: Affective Emotion

A Single Combined Session, Not Two Separate Studies Study 1 (cognitive perception, RQ1) and Study 2 (affective emotion, RQ2) were administered within the same session for every participant, rather than as separately sampled studies—for generalizability (re-contacting Study 1 participants later would retain only a self-selected, attrition-biased subset), internal validity (a delay would introduce an intervening period of ordinary activity that could confound the cognitive-affective link under investigation), and because this reflects

a Sequential Explanatory Design (Ivankova et al., 2006; Creswell and Clark, 2011), in which quantitative data establish a hypothesis’s macro-level tendency immediately followed by elaborating qualitative data. This design is also what enables the within-participant correlation analysis between cognitive accuracy, response duration, and SFF-scored affect (see “Quantitative Analysis Pipeline” below), rather than only separate-sample aggregate comparisons; since our goal is the immediate cognitive-affective response rather than its evolution over time, we judge the single-session design more rigorous here. An overview of the full pipeline, including the two points at which the LLM enters (stimulus generation and SFF scoring), is given in Appx. B (Fig. 6).

Contextual Anchors and the SFF Framework

Since pilot tests showed that describing affective responses to raw HEX swatches was cognitively taxing and abstract, we used Gemini 2.5 Flash to generate a contextual scene anchor for each target HEX code (e.g., #008080), reasoning that social emotions like Formality and Friendliness are inherently contextual. Three images were independently generated per color and one sampled uniformly at random per participant to avoid scene-specific bias; images were cropped in Adobe Photoshop for aspect ratio only, with no color alteration (nine total, Fig. 2). The SFF framework’s three dimensions draw on established pragmatics concepts: Sentiment captures affective polarity (Russell, 1980); Formality reflects social distance per politeness theory (Brown and Levinson, 1987); and Friendliness captures interpersonal warmth. The Formality/Friendliness split was informally motivated by the Korean honorific system, where *banmal* (casual) vs. *jondaemal* (honorific) speech does not by itself determine social distance—e.g., speakers routinely use formal register in public settings with otherwise close, familiar addressees, decoupling formality from interpersonal closeness (Tomioka and Cho, 2021). We treat this as an informally-grounded intuition rather than a claim validated specifically for color-affect scoring, where we are unaware of prior work; it nonetheless resonates with LLM cultural-benchmarking findings that models score differently along Hofstede-style dimensions by country of origin (e.g., Chinese-origin models scoring higher on power distance and collectivism than US-origin models; (Wang et al., 2024))—external corroboration, not direct validation.



Figure 2: The nine Gemini 2.5 Flash-generated contextual stimuli used in Study 2, three per target HEX code (rows: Blue #0000FF, Green #008000, Teal #008080). One image per color was shown to each participant, sampled uniformly at random.

LLM-based SFF Quantification We adopt a large language model (LLM) for this scoring step, rather than a classical encoder (e.g., BERT-style models) or a smaller local model (e.g., 8B-parameter open-weight models): an LLM’s pre-training corpus is substantially larger and more diverse, and its resulting embeddings are correspondingly more finely structured, internalizing a broader range of the complex socio-linguistic reality of everyday language use—implicit tone, register, and interpersonal nuance—than classical or smaller models’ more limited pretraining data and embedding capacity are able to capture. Specifically, we utilized Gemini 2.5 Flash (fixed parameters, JSON output) as a “linguistic ruler” to map participant responses into a three-dimensional SFF vector: Sentiment (S, valence), Formality (F, social distance), and Friendliness (Fr, warmth).

Quantitative Analysis Pipeline Preliminary tests showed weak correlations ($r < |0.1|$) between cognitive metrics and SFF scores, shifting focus to cultural and age factors. Our pipeline consists of: (1) **Affinity Mapping** via Euclidean distance between country-pair SFF vectors (Eq. 2); (2) **Correlation Cross-Examination** for cross-cultural baselines; and (3) **Affective Clustering** via K-Means ($K = 3$, chosen for interpretability and cross-checked via the Elbow and Silhouette methods; see Appx. D, Fig. 12) to identify linguistic cohort prototypes.

$$d(A, B) = \sqrt{(S_A - S_B)^2 + (F_A - F_B)^2 + (Fr_A - Fr_B)^2} \quad (2)$$

3.3 Data Collection and Cohort Validation

Upon completion of the survey, key demographic and control data were collected to ensure the integrity of the linguistic cohorts. We defined the primary language groups based on participants’ self-reported “primary language used in daily life.”

To ensure profound linguistic and cultural immersion, we applied a stringent dual-criteria filter (detailed in § 3.5). Participants were included in the final analysis only if their self-reported **primary language matched their current country of residence** (e.g., Korean speakers residing in South Korea). Furthermore, **a minimum residency duration of 20 years** in the respective country was required to guarantee deep-rooted cultural enculturation. Additionally, we collected data on age group to mitigate potential demographic bias in color perception. We also gathered supplementary information regarding participants’ media consumption habits (Text vs. Video and weekly consumption hours) for exploratory analysis. This allowed us to account for age-related factors while capturing a broader profile of the participants’ digital environments.

3.4 Participants, Recruitment, and Exposure Management

Participants were recruited using a strategy adapted from digital performance marketing. Initial recruitment was conducted via Prolific, a paid participant-pool platform; while this yielded a sufficient English-speaking sample, the pool of available Korean L1 speakers on Prolific proved insufficient to reach the target sample size, so recruitment was supplemented with paid social media advertising (X) and messaging-app advertising (HelloTalk), together with unpaid community posts (LinkedIn), to specifically expand the Korean-speaking participant pool. To incentivize focused participation, a gamified “personal color-perception type” feedback mechanism was provided to participants upon completion. To ensure cross-linguistic consistency, the English-speaking recruitment was calibrated to replicate the high female exposure rate observed in the Korean cohort (Korean: 91.1%; English: 92.7%). Our participant recruitment, driven by digital performance marketing, incurred a total ad spend of approximately €300 across these channels, translating to a cost per acquisition (CPA) of roughly €0.03 per participant.

Metric	Detail
Data Size	$\approx 10,000$ KR/EN Keywords
Target Model	Gemini 2.5 Flash
Ad Reach	339K+ (EN: 94K, KR: 245K)
Recruitment Spend	€295.65
CPA (Approx.)	€0.03 per response
Actual API Cost	€3.06 (Total for Oct 2025)
Total Budget	€298.71 (Recruitment + Analysis)

Table 1: Technical Specifications and Comprehensive Cost Analysis

3.5 Data Grouping, Validation, and Weighting

In total, we started with 1,838 Korean-speaking and 369 English-speaking unique individuals, alongside 57 participants from other language groups. To establish a stringent cohort of near-native speakers with deep cultural immersion, we created a dual-criteria filter: **1)** The primary language must match the country of residence (e.g., English in US, UK, or AU); and **2)** The duration of residency must be **more than 20 years**. This rigorous filtering ensured the highest quality of linguistic immersion, resulting in the final analysis cohort (**KR: 904 unique Korean L1 speakers** and **255 unique English L1 speakers**). The ETC cohort was excluded from analysis due to small sample size and heterogeneity.

After stringent data cleaning—excluding skipped, unparseable, or null (NaN) entries—**9,534 high-fidelity data points** were retained for analysis, as detailed in Table 2. This trial-level count reflects individual perception-task submissions (up to six per participant, with variation due to the platform’s stochastic, repeatable block generation described in § 3.1), and is therefore substantially larger than the unique-participant cohort sizes reported above (KR: 904, EN: 255).

Country	18–24	25–34	35–44	45+	Total
AU	24	24	24	60	132
KR	1,570	2,838	699	2,921	8,028
UK	30	110	141	659	940
US	18	52	131	233	434
Total	1,642	3,024	995	3,873	9,534

Table 2: Distribution of Study 1 Trial-Level Data Points by Age Group and Country. Counts here reflect perception-task trials, not unique participants (unique-participant cohort sizes: KR=904, EN=255).

Bias Adjustment (Oversampling and Weighting)

To mitigate inherent age-related imbalances between the cohorts, we employed a post-stratification weighting strategy. We established the existing age distribution of the Korean (KR) cohort as our target distribution. Subsequently, a statistical weighting procedure was applied to the English (EN) cohort to align its effective age distribution with this KR

target sample. The weight $w(a)$ assigned to each EN data point in age group a was calculated as:

$$w(a) = \frac{P_{\text{KR}}(a)}{P_{\text{EN}}(a)} \quad (3)$$

where $P_{\text{KR}}(a)$ and $P_{\text{EN}}(a)$ denote the proportions of age group a in the KR and EN cohorts respectively. Full details in Appx. B.2.3.

A preliminary suite of GLM tests on the original data confirmed that Age Group was highly significant in predicting error rate ($p < 0.001$; see Appx. C (Table 6 and Table 7)). This strong, unadjusted effect underscored the necessity of applying the Age-based weighting strategy to prevent demographic bias from confounding the Language main effect. Furthermore, to ensure statistical comparability in specific secondary analyses—such as the correlation between media consumption and performance—the age distribution was standardized by weighting each age group to an equal proportion of 25%. This standardization minimized any age-driven skewness, allowing for a more isolated examination of media-related variables. (e.g., media type correlation analysis).

4 Findings and Interpretation

4.1 Study 1: Cognitive Perception

To address RQ1, we analyzed the quantifiable differences in categorical perception between the KR and EN cohorts. As shown in Fig. 3 and 4, the KR cohort exhibited a highly significant performance advantage within the Grue-related spectrum. The KR cohort showed a highly significant difference in both median error rates (R_E) and response duration within the Grue spectrum (Mann-Whitney U test, $p_{\text{dur}} = 0.0036$, $p_{\text{err}} < 0.001$). Specifically, the “Yellow” category—representing the transition from Blue to Green—was associated with faster processing and significantly lower error rates for the KR cohort. This localization of the duration advantage to the Grue-related spectrum specifically parallels Winawer et al. (2007)’s classic finding that Russian speakers’ discrimination-*speed* advantage was likewise confined to trials crossing the *siniy/goluboy* lexical boundary, rather than reflecting a general processing-speed advantage. The consistently lower R_E and smaller data variance suggest a more stable and precise perceptual process in the Korean linguistic group.

The KR cohort’s superior accuracy was maintained even in categories typically associated with high cognitive load, such as the transitions to

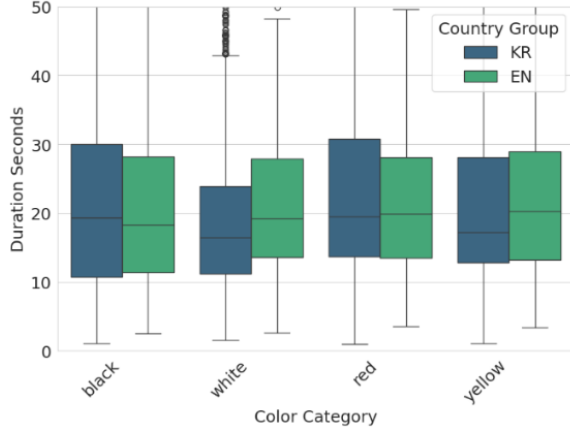


Figure 3: Comparison of Duration Seconds across color categories by language group.

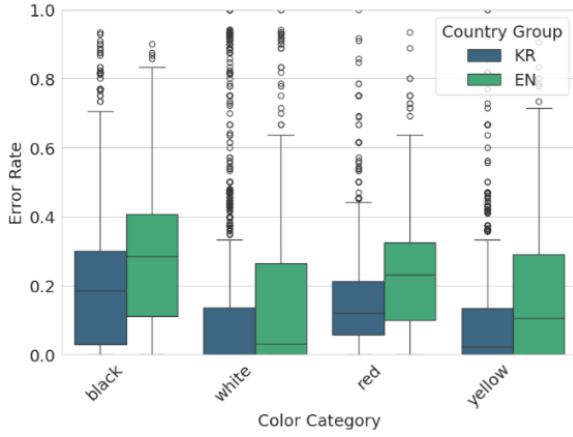


Figure 4: Error Rate (R_E) distribution by color category and language group.

Black and Red. While these darker or more distant gradients generally increased error rates across all participants, the KR cohort demonstrated significantly greater resilience and precision. This pattern is consistent with the Korean cohort’s perceptual advantage reflecting a cognitive trait associated with their specific linguistic framework, extending across the entire Grue spectrum. Specifically, the KR cohort demonstrated consistently higher accuracy (lower R_E) compared to the UK and US groups; the comparison with AU showed no statistically significant difference once age, color category, and media covariates were controlled for (the “Australia Divergence,” discussed further below). The comprehensive box plots illustrating these distributions provided in Appx. C (Fig. 8).

4.1.1 Impact of Color Categories on Estimation Difficulty

The Generalized Linear Model (GLM) and subsequent Mann-Whitney U tests revealed that color categories significantly influence cognitive performance, with a clear trend linked to the dark-

ness of the stimuli. Notably, the order of estimation difficulty observed in this study—where error rates were lowest for Yellow and White, followed by Red, and highest for Black—closely aligns with the frequency of color terms in Korean free recall tasks, which typically follow the order of Blue, Yellow, Red, Black, and White (Lee and Kim, 1996).

Luminance Sensitivity and the Grue Spectrum

Our results corroborate previous findings that Koreans exhibit higher sensitivity to subtle brightness variations within the blue spectrum compared to English speakers (Deshpande, 2015). Within the Grue (Green-Blue) spectrum specifically, the KR cohort maintained a significantly lower median error rate than the EN cohort ($p < 0.001$, Mann-Whitney U test). This Grue-specific accuracy gap is one instance of a broader, category-general pattern rather than a Grue-exclusive effect: the same language-group advantage holds in the GLM pooled across all four color categories ($\beta = 0.152$, $p = .007$; Table 8, Appx. C), whereas the covariate-adjusted estimate restricted to Grue alone does not independently reach significance ($\beta = +0.218$, $p = .085$, Appx. C), reflecting the smaller English-speaking Grue-only sample ($n = 293$ trials). We therefore attribute this pattern to the general morphological richness of the Korean color lexicon (Appx. A) rather than to a mechanism specific to the Grue transition alone.

Estimation Difficulty: From Yellow/White to Black

A universal trend emerged where estimation difficulty—measured via R_E —increased proportionally with darkness, from Yellow/White (lowest; $Coeff_{Yellow} = -0.890$, $Coeff_{White} = -0.459$; $p = 0.000$) through Red to Black (highest; $p = 0.000$). Critically, this ordering is not specific to Korean speakers: it is observable in the raw per-country medians for Korea, the US, and the UK alike, and persists in the GLM once language group and demographic covariates are controlled for. We therefore treat this luminance-driven ordering as a perceptual regularity layered underneath the language-specific KR advantage discussed above, rather than as further evidence of it: the two effects are additive, not the same effect described twice.

Age-Related Trends in Perceptual Precision

To further investigate demographic influences, we analyzed the error rate (R_E) across age groups (Fig. 5). Both cohorts exhibited statistically significant differences in R_E (Kruskal-Wallis H -test, $p < 0.001$), while no significant differences were observed

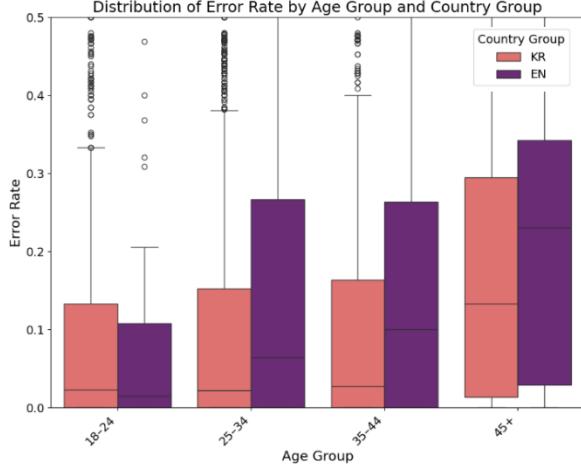


Figure 5: Error Rate (R_E) distribution by language group and age.

for response duration. Notably, the KR cohort maintained a relatively stable median error rate across the 18–44 age range, with a sharp increase occurring only in the 45+ group. In contrast, the EN cohort showed a progressive increase in median error rates with age. This plateau in the KR cohort’s performance until mid-adulthood may suggest that the cultural and linguistic reinforcement of the “Grue” category provides a cognitive buffer that stabilizes perception before age-related decline. The lack of significant age-driven variance in duration further indicates that the observed R_E trends reflect changes in perceptual precision rather than a mere trade-off in processing speed.

4.1.2 Impact of Media Consumption on Cognitive Performance

The GLM and Mann-Whitney U test confirmed Media Type (Text vs. Video) as a significant predictor of Estimate Error (R_E , $p = 0.0115$). To isolate media effects from age-related trends (as text preference increased with age), we standardized age distribution to a 25% weighted proportion. Across all cohorts, text-consumers consistently showed lower error rates, supporting the hypothesis that reading-driven analytical processing enhances cognitive precision. Given the skewed R_E distribution, we report the Median (MWU) as a more robust metric than the GLM Mean.

Language-specific analysis revealed negligible duration differences (0.06s) and no significant interaction between language and media habits regarding error rates. This suggests that media consumption does not account for the specific “Grue” advantage observed in the Korean cohort. The advantage remains robustly linked to linguistic and cultural frameworks rather than daily media

styles, necessitating further research into cultural dimension metrics.

4.1.3 Cognitive Similarity Between Countries

The resulting p -value heatmap identified two primary patterns: a stable English-speaking cluster (UK and US) with high similarity in performance metrics, and a notable cross-cultural convergence between the KR and AU cohorts specifically in accuracy, who showed no statistically significant difference in error rate despite their linguistic differences ($p = 1.000$, Bonferroni-adjusted Dunn’s test). This non-parametric result agrees with the multivariate GLM (Table 6), which likewise shows no significant KR-vs-AU contrast once age, color category, and media covariates are controlled for ($p = 0.973$), while KR differs significantly from both UK ($p = 0.028$) and US ($p = 0.039$). This accuracy-specific convergence did not extend to response duration, where KR and AU differed significantly in the non-parametric test ($p < 0.001$, Dunn’s test), though this particular contrast did not survive the equivalent multivariate adjustment (Table 7, $p = 0.055$). The heatmap (see Appx. C, Fig. 10) provides a quantitative overview.

4.2 Study 2: Affective Emotion and the Divergence

To address RQ2, we now turn from cognitive perception to the socio-linguistic and affective modeling of the Grue spectrum via the SFF framework. Beyond surface-level semantics, human language is driven by intent, implication, and situational nuances. True linguistic competence requires explicitly modeling pragmatic features—such as speech intent and audience-specific expression—rather than relying solely on distributional statistics. Our findings suggest a notable decoupling in cross-cultural psychology: Cognitive accuracy and affective expression do not inherently map onto one another. SFF similarity mapping (Appx. C, Fig. 11) shows Study 1’s alignment shifts fundamentally in emotional dimensions.

The Australia Divergence: An Illustrative Case for SFF (132 AU trials) As a secondary, illustrative case study—rather than the paper’s central evidence, given the comparatively small Australian subsample—we highlight the comparison between the Korean (KR) and Australian (AU) cohorts. We term this the “Australia Divergence,” which unfolds in two distinct stages: Accuracy Similarity ($KR \approx AU$): In the perceptual task, the AU cohort’s error rate was not statistically distinguishable

from that of the KR cohort ($p = 1.000$, Bonferroni-adjusted Dunn’s test; corroborated by the multivariate GLM, $p = 0.973$, Table 6). A Two One-Sided Test (TOST) further supports equivalence of these cohorts in error rate within a ± 0.2 -pooled-SD margin ($p = 0.003$, Appx. C, Table 10), upgrading the finding from an absence of a significant difference to positive evidence of equivalence. This established an accuracy-specific benchmark, positioning both groups within a shared “High-Accuracy” tier regarding the Grue spectrum. Notably, this equivalence was specific to accuracy: response duration diverged significantly between the two cohorts in the non-parametric test ($p < 0.001$, Dunn’s test), a pattern that foreshadows the affective divergence described next more closely than it does the accuracy result above, although this specific contrast fell just short of significance once the same demographic and task covariates were controlled for in the GLM ($p = 0.055$, Table 7). Affective Separation (KR $\neq$ AU): This separation is most pronounced when emotional responses were quantified via SFF. In the affective feature space, the AU cohort aligned exclusively with the other English-speaking groups (US, UK), moving away from the KR cohort.

The Decoupling of Perception and Expression

This divergence suggests that even when the underlying perceptual ability is shared, the socially coded rules used to express and categorize emotion may remain culturally distinct. While KR and AU participants may “see” the color boundaries with comparable precision, they appear to “feel” and “label” them through different linguistic frameworks. This finding supports the value of our methodology: the SFF framework appears to function not merely as a qualitative supplement but as a useful analytical tool for this purpose, aiming to capture culturally mediated social cues—such as Formality and Friendliness—that remain largely invisible to traditional cognitive performance metrics. The SFF framework’s ability to capture these distances is further exemplified by the qualitative segregation of semantic networks (see Appx. D, Fig. 16 for qualitative divergence in teal clusters). This decoupling carries a practical implication for culturally-aligned generation: a model optimized only for perceptual or categorical accuracy in a domain like color naming may still fail to reproduce the culturally-specific social and affective connotations attached to those categories, suggesting that alignment strategies may need to jointly model perceptual and socio-affective signal rather than treating categorical accuracy as

a sufficient proxy for cultural alignment.

SFF Quantitative Clustering and Qualitative Mapping

The SFF (Sentiment-Formality-Friendliness) framework allowed for a multidimensional quantification of affective language. By applying K-Means clustering ($K = 3$) to the SFF vectors — a choice cross-validated via the Elbow and Silhouette methods (Appx. D, Fig. 12) — we identified distinct emotional archetypes across the Blue, Green, and Teal spectrums (see Appx. D, Fig. 17). We note that this clustering separates primarily along affective valence rather than by country membership: an Adjusted Rand Index / Adjusted Mutual Information check against country label is negligible at every tested K from 2 to 8 (Appx. D, Fig. 13). The country-level affective divergence central to the Australia Divergence (§ 4.2; Appx. C, Fig. 11) instead comes from the weighted country-average SFF profiles and the pairwise distance/correlation structure between countries, not from this participant-level clustering. A complementary, qualitative case study of keyword co-occurrence for a single representative stimulus (Appx. D, Fig. 14) further illustrates lexical differences between the Korean and English networks for that specific stimulus.

5 Conclusion

This study investigated the interplay between categorical color perception and affective expression through a large-scale, cross-cultural dataset ($N = 9,534$ trial-level observations from 1,159 unique participants). Our findings are consistent with a cognitive-affective decoupling: even when perceptual precision is shared—as illustrated by the secondary “Australia Divergence” case study (132 AU trials)—the affective mapping appears shaped by culturally distinct linguistic frameworks, suggesting that dimensions like Sentiment, Formality, and Friendliness may help capture a language’s “emotional fingerprints” beyond what traditional psychophysical metrics reach. Whether this particular three-axis framing is the most advantageous one, however, remains open: future work should test alternative variables and validate the SFF scores against human judgments (see Limitations). As an exploratory, correlational study, these findings should be read as patterns warranting further confirmatory investigation rather than established causal claims.

Ethical Considerations

To ensure reproducibility and transparency, the full dataset ($N = 9,534$), Model Card, and Datasheet are publicly available at <https://osf.io/xub35/>. The source code for the experimental platform (Vibe Coding implementation, English deployment) is publicly released at https://github.com/bakeyeon/colorfish_public. The research plan and experimental platform were presented to senior researchers in the field prior to deployment, with no formal objections raised during the subsequent research period. This study was conducted as a voluntary, interest-based experiment. While a formal written consent form was not utilized, all participants were briefed on research objectives via the introductory page; voluntary submission was treated as implied consent. To protect privacy, we collected only minimal, anonymized demographic data, and sensitive information—such as participation timestamps—was deleted prior to analysis. Detailed information regarding data collection, model configurations, and bias mitigation is documented in our Model Card and Datasheet. Furthermore, contact information was provided to facilitate any research-related inquiries.

Limitations

Gender Data Omission and Exposure Bias

A primary limitation of this study is the omission of gender as a demographic variable for the initial Korean cohort ($N \approx 2,000$). Although gender data was available via the Prolific platform for subsets of the Korean ($n = 46$) and English ($n = 142$) cohorts, a technical omission regarding the Prolific PID integration prevented us from linking this data to individual survey responses. Consequently, gender-specific analyses remained precluded for the main dataset. Furthermore, ad platform analytics indicated that over 90% of initial impressions were delivered to female audiences, suggesting a potential exposure bias. While monitored sub-cohorts showed a balanced gender distribution, the lack of a direct link to the main cohort prevents us from validating the gender balance of the total sample. We addressed this by updating the data collection platform on October 24, 2025, ensuring the inclusion of gender variables for all future research phases.

Country-Level Recruitment and Exposure Bias

Beyond the gender-specific exposure bias discussed above, we cannot rule out country-level

exposure bias in recruitment. The Korean and English-speaking cohorts were recruited through the same social media advertising platform with calibrated female-exposure rates (Korean: 91.1%; English: 92.7%; §3.4), but this calibration targets only the gender axis; it does not guarantee comparability along other dimensions the platform’s ad-delivery algorithm may vary by country—e.g., socioeconomic status, education level, urbanicity, or the kind of user who self-selects into a gamified survey with performance feedback. Our age-based reweighting procedure (§3.5) corrects for compositional differences in age only, and cannot correct for these other unmeasured, potentially country-varying recruitment-channel effects. We have no specific evidence of a directional bias from this source, but we flag it as an assumption our design cannot fully verify.

Image Reproducibility and Ecological Validity

The use of AI-generated images for contextual anchors introduces a challenge for strict visual reproducibility, as generative models evolve over time. However, this was a necessary trade-off to prioritize functional and ecological validity. Measuring social emotions—specifically Formality and Friendliness—requires stimuli to be perceived within a socio-cultural framework, an environment that de-contextualized color chips cannot provide. Furthermore, while hardware variables such as screen calibration were not strictly controlled, the large-scale nature of the dataset ($N = 9,534$) suggests that individual display variances are statistically mitigated, allowing the overarching linguistic patterns to remain robust. This is consistent with broader findings that web-based, uncontrolled-environment data collection can approximate laboratory-quality results at sufficient scale (Sprouse, 2011; Crump et al., 2013); while these studies validate crowd-sourced platforms (Amazon Mechanical Turk) rather than our specific bespoke platform, they support the general principle that a large, web-based sample can compensate for reduced environmental control relative to a laboratory setting. A related confound is that, for each color, the three independently generated images vary in scene content, composition, and style (e.g., an “underwater diver” scene for one Blue-spectrum stimulus, Appx. D.2), even though each was verified to embed the same target HEX code; participants’ affective keywords may therefore respond, in part, to this scene-level semantic content rather than to the anchor color alone. We mitigate this at the aggregate level by ran-

domly sampling one of the three per-color images independently for each participant (§ 3.2, “Contextual Anchors and the SFF Framework”), so that no single scene’s idiosyncratic associations should systematically bias the sample as a whole. This does not, however, eliminate the confound at the level of any individual participant’s response—each participant’s keywords are still shaped by whichever one of the three images they happened to see—and we cannot fully decompose color-driven affect from scene-driven affect within a single generated image. We treat this as a partially mitigated, not fully resolved, limitation of our design.

Axis Collinearity in the SFF Framework

We note a measurement limitation in our LLM-based SFF quantification: Sentiment and Friendliness scores exhibit high collinearity under the current prompting scheme ($r \approx 0.97\text{--}0.98$), suggesting these two axes may not be fully separable using Gemini 2.5 Flash as a scoring instrument. Formality, by contrast, is comparatively independent of both ($r \approx 0.02\text{--}0.08$), indicating that the collinearity is specific to the Sentiment-Friendliness pair rather than a symptom of the three-dimensional framework collapsing more generally. Disentangling Sentiment from Friendliness—potentially through human annotation or refined prompting—remains an important direction for future work. Separately, we assessed whether the scoring instrument itself is stable across the LLM’s generation temperature (Appx. C, “SFF Temperature-Robustness Check”); we find no statistically significant instability once corrected for multiple comparisons, though instability specific to low-magnitude (near-neutral) keyword scores is noted there as well. More fundamentally, we have not established that Sentiment, Formality, and Friendliness are the most advantageous three variables for capturing culturally-mediated socio-affective cues in this domain, as opposed to some alternative or additional set of dimensions; future work should both explore this variable-selection question directly and pursue the human validation noted above, rather than treating the present three-axis framing as settled.

Absence of Human Validation for SFF Construct Validity

The SFF scores reported throughout this paper are derived entirely from an LLM-based scoring procedure (Gemini 2.5 Flash) and have not been validated against human raters; we do not report inter-rater agreement or any other measure of alignment be-

tween the LLM’s scores and human judgments of Sentiment, Formality, or Friendliness. This is a meaningful limitation for a construct that is ultimately about human affective and socio-linguistic experience. A full human-annotation study was not feasible within the scope of this work: it requires an annotation protocol, ethics/IRB approval, and additional funding beyond what was available for the present investigation. As a partial, computational check in the meantime, we confirmed that the scoring instrument itself is stable: re-scoring the same keywords across four generation temperatures produced highly consistent SFF vectors (Appx. C, “SFF Temperature-Robustness Check”; median cosine similarity ≈ 0.99 , no significant instability after correction for multiple comparisons). This does not substitute for human validation, but it does show the instrument is not producing arbitrary or noisy scores. We name this gap explicitly rather than leaving it as an implicit omission, and a dedicated human-in-the-loop validation study, including inter-rater reliability metrics between human annotation and the LLM’s scores, is planned as future work.

Human Verification of the Supplementary Korean-English Derivation Comparison

The English translations underlying the free-translation-vs-morphological-derivation comparison in Appendix A.1 (Table 3) were generated solely by Gemini, without verification by a human English linguist; a linguist may identify additional valid morphologically-derived English forms that Gemini did not produce, which would narrow the observed gap. We nonetheless consider this comparison useful as an approximate, illustrative estimate—given Gemini’s training on a very large English corpus—rather than an exhaustive or definitive count. This supplementary analysis is not part of this paper’s own experimental design or core claims. Furthermore, this illustrative list is not intended as an exhaustive derivational inventory, but rather as a demonstration of the well-attested sparsity of English color-term derivation. Prior work on English morphology (Matschi, 2004) similarly shows that English color adjectives generate only a small number of productive derivatives. Although Matschi’s analysis is limited to English and does not involve cross-linguistic comparison, it provides an independent empirical basis for the sparsity observed here. Additional verification by native English linguists could further refine the inventory and potentially narrow the observed gap; however, given the substantial magnitude of the

difference, it is unlikely to overturn the qualitative cross-linguistic contrast with Korean.

Feature-Space Sufficiency for Cultural Clustering

As a robustness check beyond the $K = 3$ Elbow/Silhouette analysis reported in Appx. D, we computed the Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI) between the K-means cluster assignment and participants' country label for every K from 2 to 8. Agreement remained negligible at every K ($\text{ARI} \leq 0.006$, $\text{AMI} \leq 0.017$, on a scale where even weak agreement is conventionally ~ 0.1 – 0.2), with no cluster count standing out — a K -independent finding, not an artifact of the specific $K = 3$ used elsewhere in this paper. Whether this reflects a genuine limitation of the three-axis SFF space, or simply high inter-individual heterogeneity in affective response, remains an open question. Addressing this would require an Arousal (VAD) dimension scored via the same LLM-as-rater procedure used for SFF; lexicon-based VAD resources (e.g., NRC-VAD) are English-only and would systematically fail to capture the Korean-language affective vocabulary that is central to this study's cultural claims, making them unsuitable for this comparison. This is planned as a follow-up study using consistent LLM-based scoring across languages.

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making them unsuitable for this comparison. This is planned as a follow-up study using consistent LLM-based scoring across languages.

Effect Size and Equivalence Testing

The cross-cultural comparisons reported here (e.g., the Australia Divergence) rely primarily on significance testing (Mann-Whitney U , Kruskal-Wallis, Dunn's post-hoc test); we supplement these with standardized effect sizes and a formal equivalence test (Appx. C, Table 9 and Table 10). Cohen's d for the KR-vs-English-speaking-cohort error-rate comparison is $d = 0.174$ (rank-biserial $r = 0.211$), with $d = 0.200$ and $d = 0.181$ for KR-vs-UK and KR-vs-US individually — consistently small by conventional benchmarks despite very low p -values in several cases, reflecting the large sample size (n in the thousands) rather than a dramatic effect; this is worth stating plainly; statistical significance here does not imply a large practical effect, which is consistent with this paper's exploratory framing rather than in tension with it. In natural units, the country-group GLM coefficient corresponds to the English-speaking cohort's error rate averaging $1.16\times$ that of the Korean cohort (equivalently, a 14% lower error rate for Korean speakers), holding other covariates fixed; this pooled estimate, rather than a Grue-only one, is the better-powered test of the language effect (Appx. C, "Grue-Only GLM Specification"). Separately, because R_E is exactly zero on trials where the participant's estimate exactly matches the true block count (34.3% of trials), it lies outside the strictly-positive support the Gamma family (used for both GLMs) formally assumes; we verified this technical mismatch does not drive the reported language effect via a Tweedie-family robustness check (Appx. C, "Distributional-Family Robustness Check"), under which the effect's sign, magnitude, and significance remain stable. For the "Australia Divergence" specifically, a Two One-Sided Test (TOST, ± 0.2 -pooled-SD margin) supports equivalence of KR and AU in error rate within this margin ($p = 0.003$), corroborated by a nonparametric bootstrap check on the median difference. The same test does not establish equivalence for response duration ($p = 0.625$), consistent with §4.1.3's finding that duration, unlike accuracy, diverges between these cohorts. The equivalence margin is a defensible convention (Cohen's $d = 0.2$, the conventional small-effect threshold) rather than one derived from the data, and a stricter or looser margin could change the conclusion; future work should explore

sensitivity to this choice.

Model Volatility and the “Moving Target” Problem

This study proposes the SFF framework by relying on the inherent conceptual understanding of LLMs to quantify “Formality” and “Friendliness” without explicit rubrics. The conceptual independence and construct validity of these axes therefore remain to be verified. This “LLM-as-ruler” approach has a further, more fundamental limitation: benchmarking work has found that LLMs reflect the cultural values of their country of origin only to a degree, without fully replicating human value systems (Karinshak et al., 2024). We take this as grounds not to abandon the LLM-as-rater approach outright, but to treat any single LLM’s scores as an approximation of cultural affect rather than ground truth—motivating future work toward a validated, human-annotated “golden dataset” and a more principled scoring framework, rather than relying on a single model’s outputs alone. Furthermore, using LLMs as measurement instruments highlights a critical challenge: model reproducibility.

Our results are tied to the Gemini 2.5 Flash model version from late 2024. The continuous updates of these models act as “moving goalposts,” threatening the long-term validation of LLM-based social science. We urge model providers—Google, Meta, OpenAI, and Anthropic—to provide persistent “academic snapshots” via dedicated APIs. Establishing a reliable versioning system, such as five-year archival keyframes, will be the most critical scientific legacy this generation of AI developers bequeaths to future researchers.

Generalization and Representation

While this study evaluates the pragmatic gap using Gemini, we acknowledge that the results may not be fully generalizable to all Large Language Models (LLMs). Different architectures and training corpora may lead to varying degrees of cultural sensitivity. However, as Gemini is recognized for its robust multilingual capabilities, our findings serve as a meaningful indicator of how even advanced models struggle with culturally-nuanced color-affective associations. Future research should extend this evaluation to a broader range of models to validate the consistency of these gaps.

Neuroscientific Validation of the Usage-Salience Hypothesis

Our findings suggest that usage salience—the frequency and intensity of a color term’s use—is a more powerful predictor of cognitive effects than structural complexity. While these behavioral results are consistent with linguistic usage intensity being associated with perceptual accuracy within the Korean “Grue” spectrum, the study lacks direct neuroscientific evidence. Future research should integrate neuroscientific methods, such as fMRI, to validate these behavioral correlations and provide direct evidence of the neural mechanisms underlying the Korean derivational system’s effect on perception.

Affective Load and Cognitive Performance

Building on the methodology of using color overlays to modulate emotional state (Yoon and Yang, 2016), we propose a series of dedicated cognitive experiments to empirically verify the findings regarding selective processing and cognitive load observed in Study 1. Specifically, we aim to test whether colors that induce low cognitive friction (e.g., Yellow and White) lead to measurably faster and more frequent positive behavioral outcomes compared to high-load colors (e.g., Black). This phase aims to isolate the direct impact of culturally mediated cognitive load on subsequent low-effort actions, providing practical implications for design and communication.

Linguistic Expansion and Cultural Assimilation

To further generalize these findings, future research must expand the cohort to include other East Asian languages, particularly Japanese (JP). Given that older Korean speakers utilize Japanese-influenced vocabulary for color more frequently (Tyson, 1998), a comparative analysis of KR, JP, and EN cohorts is essential to isolate the specific influence of native Korean derivational systems from shared regional linguistic contexts. This will provide insights into how varying rates of cultural assimilation and English adoption impact the erosion or preservation of traditional linguistic color boundaries across East Asia.

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Final Note on References: For academic works originally published in Korean where no official English translation of the title exists, the titles have been translated by the authors for clarity.

Appendix

Appendix A: Cultural and Linguistic Background of Korean Color Perception

This appendix provides a brief summary of the foundational concepts derived from historical Korean linguistic studies which informed our survey design and hypothesis on color perception and emotional association among Korean speakers.

Traditional definitions of primary colors often rely on Western physical models such as RGB or CMY(K). Korean color vocabulary to five terms (*Obangsaek*): **white, red, yellow, black, and blue**. The Korean perception of color has evolved beyond “visible colors” to “conceptualized colors” (Lee, 2001). This approach is deeply rooted in the Yin-Yang and Five Elements philosophy, which maps the five cardinal colors to **diverse cultural contexts**. In this framework, Korean colors extend beyond visual stimuli to signify directions, flavors, and other socio-cultural attributes (Kang, 2009).

In a Korean color-overlay experiment (Yoon and Yang, 2016), cognitive performance significantly decreased when subjects in a positive emotional state viewed red overlays, whereas green overlays under a negative emotional state significantly enhanced performance. Moreover, the application of color overlays appeared to mitigate the influence of initial emotional states on cognition.

A.1 Grue Culture in Korea

Significant naming disagreements exist between Korean and English speakers, particularly within the green category has been observed. The experiment also exhibited a unique pattern in the blue spectrum between light and dark regions. (Deshpande, 2015). Color naming practices also reveal a generational shift; while younger Koreans (ages 18–24) show higher consistency by preferring native Korean terms, older generations (ages 47–85) frequently utilize Sino-Korean (*Hanja*) or Japanese loanwords, thereby maintaining the traditional “grue” category (Tyson, 1998).

A quintessential example of the grue category in Korea is the word *pureuda*, which encompasses both green and blue. Consider the following examples:

1. “*Haneul-i pureuda*” (The sky is blue).
2. “*Soop-i pureuda*” (The forest is green).

Another compelling instance is the “green” traffic light, which Koreans call *paran* (typically meaning

”blue”), suggesting that the Korean color prototype for this spectrum remains anchored in a broader ”grue” category, where blue acts as a primary lexical anchor. A similar phenomenon is observed in Japanese, where *Ao* (blue) is used for the green traffic light (*Ao Shingou*), suggesting a shared regional or linguistic influence between these neighboring cultures. The etymological origins further reinforce **blue as the anchor of the Korean Grue spectrum**; while the word for green, *cho-rok*, is a Sino-Korean loanword, the word for blue, *parang*, is a native Korean term with no Chinese character equivalent. Furthermore, the root of *parang* exhibits exceptional grammatical flexibility, allowing for highly productive morphological variations that denote subtle nuances (Lee, 2001).

This structural depth is supported by a large-scale analysis of over 42,000 free-recall responses (Lee and Kim, 1996), which revealed that the **Grue-spectrum** (rooted in *pureuda*) is the most actively utilized category in the Korean color lexicon.

The structural depth of the Korean color lexicon can be further understood through the lens of **morphosyntactic trade-offs**: linguistic functions originally realized through syntactic structures can, in some languages, shift over time toward inflectional morphology (Haspelmath and Sims, 2013). We note that Haspelmath and Sims (2013) establish only the existence of such trade-offs in general; a simple inverse relation between morphology and syntax is not claimed to hold universally across languages, so we treat this as a lens for interpretation rather than a settled mechanism we can assume applies here.

Building tentatively on Haspelmath and Sims (2013)’s framework of morphology-syntax trade-offs, we suggest that this pattern may reflect not merely superficial formal variation, but possibly a deeper functional strategy—a framing that goes beyond what Haspelmath and Sims (2013) themselves establish, and one for which further psycholinguistic evidence would be needed. While English relies on complex syntactic constructions (e.g., phrases with multiple modifiers) to describe nuances (Matschi, 2004), Korean utilizes productive morphological transformations (Lee and Kim, 1996), compressing nuanced semantic information into a single word via affixes. We suggest this pattern is at least broadly consistent with Korean’s morphologically rich color vocabulary offering some processing advantage by reducing working-memory burden and lessening reliance on hierarchical syntactic parsing—though whether this translates

Corpus / Translation	# Distinct Forms
Korean morphological	419
English, free translation	201
English, morphological	20

Table 3: Translating a 419-term Korean color-vocabulary corpus into English via Gemini under two conditions: free translation (any natural construction) vs. translation constrained to morphological derivation only. The roughly tenfold gap suggests English relies predominantly on syntactic, not morphological, means to encode this variation.

into a genuine cognitive effect remains a hypothesis rather than a demonstrated one. Given this intensity of lexical use, this study focuses on an in-depth exploration of the Grue spectrum in Korean.

We tested the syntactic-vs-morphological half of this claim directly, as a supplementary check not part of this paper’s own experimental design: translating the same 419 Korean derivational variants (built from terms in Lee and Kim (1996)’s free-recall survey) into English via Gemini under two conditions—an unconstrained free translation permitting any natural construction, including syntactic modifiers and compounds (e.g., *sallow*, *coal-black*), versus a translation constrained to root-internal-modification or affixal derivation only (e.g., *yellowish*, *blackish*)—yields a striking gap (Table 3). Free translation produces roughly ten times as many distinct English forms as the morphologically-constrained translation, consistent with English encoding this fine-grained color variation predominantly through syntactic rather than morphological means, in contrast to Korean’s reliance on morphology for the same 419 variants. This gap is most extreme for yellow, where the single morphologically-derived English form *yellowish* absorbs well over 100 distinct Korean variants on its own. We note that this English-side inventory was derived by translating from Korean rather than independently compiled by a native-English-speaking linguist, so it should be read as a lower bound on English’s own productivity rather than a precise, independently-collected estimate; the qualitative direction of the gap is nonetheless unlikely to reverse under independent collection, given how few of the free-translation forms (e.g., *sallow*, *coal-black*) are themselves morphologically derived from a shared root.

One candidate mechanism for such an effect can be situated within the Label-Feedback Hypothesis (Lupyan, 2012), which proposes that linguistic labels can exert rapid, top-down modulation on perceptual representations by activating category-

specific prior information. Empirical work under this hypothesis reports that even basic visual tasks—color identification, visual search, simple detection—can be facilitated when a relevant label is activated, with effects appearing within a few hundred milliseconds, suggestive of dynamic feedback between the lexical and perceptual systems. If Korean color adjectives do encode hue, saturation, and emotional tone in a single morphologically compact form, it is possible that such labels could function as a comparatively strong perceptual cue, potentially reinforcing category boundaries and facilitating faster color identification—though we offer this only as a plausible extension of the hypothesis to our specific case, not as a claim we test directly in this study.

Specifically, as highlighted by Lee(2001), the Korean perception of color moved beyond 'seeing color' to 'thinking color,' becoming highly conceptualized. This framework emphasizes that colors are interpreted through the lens of cultural context and philosophical principles (e.g., the Yin-Yang and Five Elements). This perspective formed the foundational basis for our study's unique survey design.

A.2 Foundational Concepts and Five Key Characteristics of Korean Color Consciousness

Korean color consciousness is deeply rooted in philosophical and cultural history, evolving from basic visual sensation to 'conceptualized thought'. Our analysis of traditional Korean color terms reveals that pure Korean color names (like *parang*, *ppalgang*, *norang*) possess a unique linguistic structure allowing for complete functionality as nouns, adjectives, and adverbs through simple affixation, demonstrating a sophisticated, language-dependent cognitive framework. Lee emphasized that "*Korean color consciousness has evolved beyond visible colors to become conceptualized as thoughtful colors.*" This suggests that Koreans' perception of color goes beyond simply seeing colors, but has a deeper cultural depth.

This overall consciousness of color among Koreans, as documented in classical texts, is structured by the following five principles:

1. **Meaningfulness:** Colors are imbued with philosophical meaning based on the Yin-Yang and Five Elements theory (*Obang-wi*). This dictates color use in ceremonies, official dress, and architecture.
2. **Purity:** Emphasis on the fundamental state,

favoring naturalism and the preference for **White**. This is reflected in the simplicity of commoners' clothing and traditional arts.

3. **Contrast:** A focus on **cyclical contrast** based on mutual generation and subjugation principles. This is often realized through complementary and brightness contrast (e.g., Korean national flag, traditional formal wear).
4. **Seclusion:** A desire to harmonize with nature leads to the use of **low-chroma natural dyes** and muted, adjacent color contrasts, visible in traditional architecture.
5. **Humor/Wit:** Incorporating optimistic and humorous elements into daily life and folk arts (e.g., folk paintings, mask dances), based on the **Pung-ryu** philosophy.

A.3 Samples for Cultural Color Context in Korean

Blue corresponds to the east, wood and life. It symbolises dynamic energy of beginnings, growth and expansion.

Azure / Blue (Wood)

Located in the **East** and guarded by the **Azure Dragon**. It represents **Spring**, the virtue of **Benevolence**, and the **Liver**. Associated with the eyes, a sour taste, and the sound *Gak (Mi)*.

Red (Fire)

Located in the **South** and guarded by the **Vermillion Bird**. It represents **Summer**, the virtue of **Propriety**, and the **Heart**. Associated with the tongue, a bitter taste, and the sound *Chi (Sol)*.

Yellow (Earth)

Located in the **Center** (no specific guardian). It represents the **Four Seasons**, the virtue of **Trust**, and the **Spleen/Pancreas**. Associated with the mouth, a sweet taste, and the sound *Gung (Do)*.

White (Metal)

Located in the **West** and guarded by the **White Tiger**. It represents **Autumn**, the virtue of **Righteousness**, and the **Lungs**. Associated with the nose, an acrid/spicy taste, and the sound *Sang (Re)*.

Black (Water)

Located in the **North** and guarded by the **Black**

Tortoise. It represents **Winter**, the virtue of **Wisdom**, and the **Kidneys**. Associated with the ears, a salty taste, and the sound *Woo (Ra)*.

Note: Cultural dimensions adapted from (Kang, 2009).

Below are key samples of Blue, the Korean equivalent of "Pureuda."

- 숲이 푸르다 *Soop- i pureuda*.
- The forest is green.
- 푸른 봄의 기운 *Pureun bom-ui giwoon*.
- The grue energy of spring.
- 동방의 푸른 기운 *Dongbang-ui pureun giwoon*.
- The blue energy of the east.
- 동방의 수호신 청룡 *Dongbang-ui suhoshin Cheong-ryong*.
- Azure Dragon, Guardian of the East.

Additional Information: Blue is the Buddhist symbol of wisdom and compassion. / The concept of Blue/Green being associated with the Liver/Wood element is a core tenet of Traditional Korean Medicine and the Five Elements theory.

Appendix B: Technical Details and Methodology

B.1 Stimuli Generation and Experimental Procedure and Statistical Analysis Strategy

For the emotion-mapping task, all visual stimuli were generated through a collaborative process involving Large Language Models and professional graphic software. Specifically, Gemini 2.5 Flash was utilized to generate three distinct images for each designated color code (e.g., #0000FF for blue, #008000 for green, #008080 for teal/mint). To ensure rigorous experimental control, these AI-generated images were cropped in Adobe Photoshop for aspect ratio only, with no color alteration. Participants were presented with these stimuli in a randomized sequence and prompted to provide open-ended keywords reflecting their immediate emotional and sensory associations. These qualitative responses were subsequently quantified using the SFF framework. This dual-track data collection ensures that both the pre-linguistic thought and structured socio-cultural sentiments are captured.

While this generative approach introduces a reproducibility challenge—as exact image replication in evolving AI models is difficult—we prioritized the functional validity of the SFF framework over stimulus-level replication. In total, nine images were generated (three for each primary color).

Multidimensional Quantification (SFF) Each keyword was transformed into a three-dimensional vector (S, F, Fr) representing:

- Sentiment (S): The emotional valence (positive to negative).
- Formality (F): The perceived social distance or degree of politeness.
- Friendliness (Fr): The level of warmth or approachability.

B.1.1 Robustness check

Data Preservation Philosophy: We avoided external normalization because forcing a transformation on human-generated data risks losing subtle, non-linear insights inherent in the original distribution.

Data Nature The primary metric, R_E (Normalized Error Score), follows a Non-Normal Distribution (Non-Standard Normal). Requires caution; standard parametric tests are insufficient.

Primary Analysis Generalized Linear Model (GLM) with Age-based Weighting for checking significant variables. Compares the two language groups (KR vs. EN) while controlling for confounds (Age, Media Type).

Robustness Check (Non-Parametric) Mann-Whitney U Test (for group comparison) and Kruskal-Wallis H Test (for factor effects). Confirms the GLM findings and validates effects on median values, independent of distributional assumptions. The highly significant result of the Mann-Whitney U Test ($p < 0.0001$) confirms that there is a genuine perceptual advantage in the central tendency (median) of the Korean population. Because this finding is independent of any distributional assumptions and less sensitive to outliers, we trust the non-parametric finding as a strong confirmation of the Korean cohort's superior perceptual process, visually confirmed in our initial bar graphs.

B.1.2 Initial Global Participation

Initial Global Scope Participants recruited from 25 countries (excluding test/NaN).

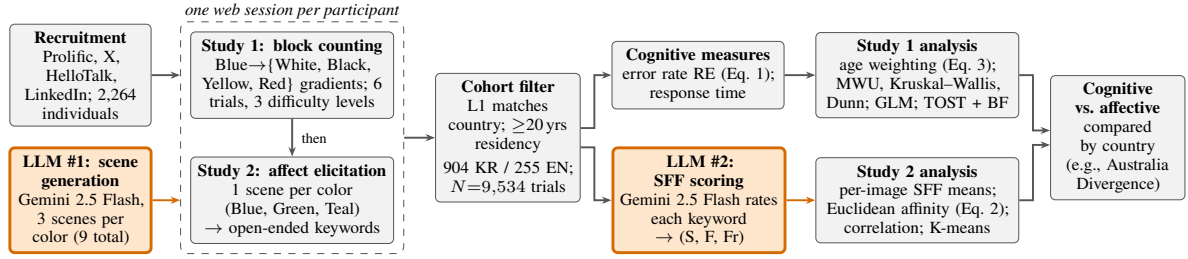


Figure 6: Methods overview. Each participant completed Study 1 (block counting) and Study 2 (affect keywords) in one web session. The LLM enters the pipeline twice (orange): generating the Study 2 scene stimuli and scoring each keyword into an SFF vector. It plays no role in Study 1.

Top Participants by Data Points Volume KR (10841), UK (1115), US (527), DE (264), AU (198), JP (42), CN (36), etc. Although we had a promising initial number DPs from Germany, after strictly applying the Dual-Criteria Filter (L1 matching DE, 20+ years residency), the remaining number was simply too small to form a robust, independent analysis cohort that could be reliably weighted and compared with the Korean and English groups. We had to focus our limited resources on the four countries that successfully passed our stringent filter.

Dual-Criteria Filter Applied L1 matches Country of Residence and Residency over 20 years. Answers that did not make logical sense were removed (e.g., respondents who responded that they had resided for more than 20 years but indicated that they were under 18 years old).

Final Analysis Cohorts KR (Korea), US, UK, and AU (Australia) remained for analysis.

Outcome Focusing analysis from broad global scope to 4 high-fidelity, culturally-immersed country cohorts. This initial, unfiltered 25-country dataset is not publicly released; only the final, dual-criteria-filtered dataset (`anonymized_df.json`, $N = 9,534$) described above is released on OSF.

B.1.3 Variable names

Released Dataset File: `anonymized_df.json`

- To prevent users from being identified by the time they submitted their data, we removed the `submitted_at` column and replaced it with `user_id`.

The names of the variables included in the updated "`anonymized_df.json`" are as follows:

- `age_group`: User Age Group.
- `country`: User's country of residence.

- `years_in_country`: The period of time the user has lived in the country of their current residence.
- `primary_language`: User's first language.
- `media_type`: User's media consumption type (text/video)
- `media_kind`: Name of media most frequently consumed by the user (subjective response)
- `media_hours_per_week`: Time users spend consuming media (weekly)
- `trial_number`: Numbering of the problem (1-6) indicating which problem it was out of 6.
- `actual_blocks`: The actual number of blocks the user had to guess.
- `user_estimate`: The number of blocks estimated by the user.
- `duration_seconds`: The time a user spends submitting a task.
- `difficulty_level`: Difficulty of the problem (1-3 levels, easy to hard).
- `color_category`: A color category that has a gradient from blue to another color (e.g. yellow = blue to green).
- `original_df`: Whether the original survey the user responded to was in Korean or English.
- `Group`: Group tags by language classified for analysis.
- `weight`: Primary weights to balance age groups across different language groups.
- `avg_duration_per_participant_x`: A variable created to calculate the average time it took one person to solve the entire task.

- *avg_duration_per_participant_y*: A variable created to calculate the average time it took one person to solve the entire task.
- *normalized_avg_duration*: Average time taken by one person to complete all tasks.
- *error_amount*: The number of blocks the user had to guess minus the user's estimated number of blocks.
- *error_rate*: Take the absolute value of the error amount, divide it by the actual number of blocks, and then divide it by the difficulty (1-3).
- *country_group*: Grouping for language-specific comparison (KR/Other=EN).
- *blue_emotion_keywords*: Keywords for blue images seen by the user.
- *green_emotion_keywords*: Keywords for green images seen by the user.
- *teal_emotion_keywords*: Keywords for teal images seen by the user.
- *shown_blue_image*: The file name of the blue image seen by the user.
- *shown_green_image*: The file name of the green image seen by the user.
- *shown_teal_image*: The file name of the teal image seen by the user.
- *blue_scores*: SFF scores (tuples) entered by the user for the blue image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *green_scores*: SFF scores (tuples) entered by the user for the green image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *teal_scores*: SFF scores (tuples) entered by the user for the teal image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *avg_sentiment*: Average sentiment score of all keywords entered by the user.
- *avg_formality*: Average formality score of all keywords entered by the user.
- *avg_friendliness*: Average friendliness score of all keywords entered by the user.
- *blue_sentiment*: Sentiment score entered by the user for the blue image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *blue_formality*: Formality score entered by the user for the blue image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *blue_friendliness*: Friendliness score entered by the user for the blue image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *green_sentiment*: Sentiment score entered by the user for the green image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *green_formality*: Formality score entered by the user for the green image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *green_friendliness*: Friendliness score entered by the user for the green image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *teal_sentiment*: Sentiment score entered by the user for the teal image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *teal_formality*: Formality score entered by the user for the teal image. If a user entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.
- *teal_friendliness*: Friendliness score entered by the user for the teal image. If a user

entered multiple keywords, the score is calculated for each keyword and then averaged to create the score.

B.1.4 Technical Details for API usage

The analysis is highly cost-efficient due to the use of the Gemini 2.5 Flash model and aggressive token reduction.

Metric	Detail
Data Size	≈ 10,000 Korean/English Keywords
Target Model	Gemini 2.5 Flash (gemini-2.5-flash-preview-09-2025)
Total Est. Tokens	1,100,000 tokens (Input: ≈ 700k, Output: ≈ 400k)
Actual Cost (Google Cloud)	€3.06 (Total in EUR for Oct 2025)
Cost Efficiency	Extremely Cost-Efficient; achieved via optimized batch processing and aggressive token reduction strategy.
Details	Temperature: 0.9 Top-p: 1.0 max.tokens: 2048 Seed: None Hardware: Google Colab (CPU runtime) Python: 3.12.12(main, Oct 10 2025, 08:52) google-generativeai: 0.8.5 Libraries: pandas: 2.2.2, numpy: 2.0.2, matplotlib: 3.10.0, seaborn: 0.13.2, statsmodels: 0.14.5

Table 4: Technical Specifications and Cost Analysis of LLM-based SFF Quantification

B.2. Detailed Data Processing and Analysis Framework

This section covers the explanation of the advanced analysis process.

B.2.1 Analysis Strategy for Color Perception Data

For the color perception data, we first calculated a normalized Estimate Error Score (R_E) for each trial. This score represents the participant’s **relative error rate** weighted by the task’s difficulty. It was calculated by taking the absolute difference between the actual block count (B_a) and the user’s estimate (E_u), and then dividing this by the product of the B_a and the difficulty level (L_d).

$$R_E = \frac{|B_a - E_u|}{B_a \times L_d} \quad (4)$$

To ensure that the observed performance gaps were not merely artifacts of demographic distribution, we employed a Generalized Linear Model (GLM) using country-specific age weights, with age group, color category, media type, and the four-level *country* variable as predictors (Table 6, Table 7). Note for users of the released

dataset: *country_group* (binary KR/Other) and *country* (four-level) are exactly collinear ($country_group[Other] = 1 - country[KR]$), so including both terms simultaneously produces a rank-deficient design matrix and non-identifiable country coefficients; we use only the four-level *country* term here. In this model, the Country effect remains highly significant for error rate, both in a supplementary pooled KR-vs-English-speaking-cohort specification ($\beta = 0.152$, $p = 0.0066$) and specifically against the UK and US individually (KR-referenced contrasts: $p = 0.028$ and $p = 0.039$, respectively), but not against AU ($p = 0.973$). For response duration, by contrast, the Country effect does not reach significance once age, color category, media type, block count, and error rate are jointly controlled for, either in the pooled comparison ($\beta = 0.008$, $p = 0.701$) or in any individual pairwise contrast with KR (all $p > 0.05$). This indicates that the raw, unadjusted duration gap discussed below is attributable at least in part to these confounding factors rather than reflecting a country effect that is robust to adjustment. As a robustness check on the choice of reference category, we additionally refit both models with UK and then US (instead of AU) as the baseline; because the corrected model is full-rank, this is a pure reparameterization, and all nine pairwise country contrasts (both metrics) are numerically identical regardless of which country anchors the model. This confirms that the KR-vs-UK/US significance paired with KR-vs-AU non-significance is not an artifact of AU’s small sample size (132 AU trials) serving as the reference category.

Prior to these adjustments, a suite of non-parametric tests on the original, unweighted data confirmed that Country exerted a significant influence on all performance metrics (Kruskal-Wallis H-tests, $p < 0.001$). However, the impact of Age Group was found to be metric-specific: while it significantly influenced R_E ($p < 0.001$), its effect on response duration was not statistically significant in either the KR ($p = 0.126$) or Other ($p = 0.592$) cohorts (detailed in Appendix C).

Our statistical approach centers on a Generalized Linear Model (GLM) using country specific weights to isolate the linguistic effect from demographic imbalances. We also utilized non-parametric tests—specifically Kruskal-Wallis H-tests and Mann-Whitney U tests—to validate the robustness of the performance gaps across different age and language cohorts.

B.2.2 Analysis Strategy for Color Emotion Data

Data Cleaning and Standardization Before all statistical analysis, any trials containing missing values (NaN) for key variables and those with response times exceeding 300 seconds were excluded from the dataset. The 'primary language' variable was subject to a **manual coding process** wherein semantically equivalent responses (e.g., 'Korean', '한국어', '모국어') were grouped into the primary categories (KR, EN). **For participants who listed multiple languages, the first language listed was designated as their primary language for analysis.** Similarly, 'country of residence' was manually standardized to its ISO 3166-1 code (e.g., 'US', 'UK', 'AU').

Data Preprocessing and Refinement The collected keywords underwent a two-stage refinement process: (1) exclusion of null entries and (2) lexical segmentation, where multiple keywords entered for a single image (e.g., "cold, solid") were treated as individual lexical items and averaged into that image's Sentiment/Formality/Friendliness score. These per-image SFF averages are merged onto the Study 1 trial-level table for joint analysis. The resulting $N = 9,534$ reported in Table 2 is therefore the Study 1 perception-trial count, not an independent count of keyword-level observations: a given participant's averaged SFF scores are repeated across that participant's (up to six) trial rows.

B.2.3 Age Group Weight Adjustment

The reason we did not aim for a perfectly 'even' age distribution across all groups is twofold:

1. **Respecting the Target Distribution:** We were specifically interested in the perceptual and linguistic performance of the Korean cohort as our primary target. Instead of artificially creating a flat, generic distribution, we adjusted the minority EN cohort to reflect the observed KR target distribution ($P_{KR}(a)P_{KR}(a)$) using the ratio formula.
2. **Focus on Correlation/Effect:** Crucially, our main analysis used a Generalized Linear Model (GLM), which is designed to analyze correlations and effects while controlling for specified variables (like Age). It does not require a perfectly flat distribution; it only requires the distributions of the two comparison groups (KR vs. EN) to be aligned to prevent bias. Our weighting successfully achieved this necessary alignment.

The weight $w(a)$ assigned to each EN data point in age group a was calculated as the ratio of the target (KR) and source (EN) proportions:

$$w(a) = \frac{P_{KR}(a)}{P_{EN}(a)} \quad (5)$$

This formula ensured that the weights of the English dataset were specifically adjusted based on the Korean dataset to align the effective age distributions. The visualizations of the adjusted distributions confirm this successful alignment: the KR and EN (Weighted) distributions are now comparable, allowing us to proceed with a statistically sound, unbiased comparison.

B.2.4 Optimized Prompt Strategy (Token Reduction)

The original prompt was refined to remove non-essential UI/logging instructions, reducing input tokens while maintaining the core analytic requirements. The system is instructed to act as a linguistic expert and output data in a strict JSON format for easy post-processing.

Target Model: Gemini 2.5 Flash (for cost efficiency and speed)

Optimized Prompt Template (Input to API):

```
prompt = f"""For each word or phrase in the
→ following list, perform a sentiment
→ analysis.
If a single item contains multiple
→ words/phrases separated by commas, spaces,
→ or other delimiters (e.g., 'cold, solid'),
treat each separated element as an individual
→ item and provide a score for each of them.
```

```
Provide a sentiment score from -1.0 to 1.0.
→ Also, determine the formality of the word
→ on a scale from -1.0 (informal) to 1.0
→ (formal),
and the friendliness of the word on a scale
→ from -1.0 (unfriendly) to 1.0 (friendly).
```

```
Provide the output as a single JSON object
→ (enclosed in {{ and }}) where the key is
→ the word/phrase and the value is an object
containing the three scores. Ensure all
→ individual scores are grouped within this
→ single JSON object.
Do not include any text, notes, or
→ explanations outside of the JSON object.
```

For example:

```
{{
  "cold": {{
    "sentiment": -0.4,
    "formality": 0.2,
    "friendliness": -0.7
  }},
  "solid": {{
    "sentiment": 0.1,
    "formality": 0.5,
    "friendliness": 0.3
  }}
}}
```

```

    }}
}}

The list of words is:
↪ {json.dumps(chunk_1)}.""

```

Appendix C: Supplementary Statistical Analysis

This section provides supplementary statistical analysis details for the findings reported in the main body of the paper.

Reproducibility: Software Environment All analyses in this appendix (GLM regressions, effect sizes, TOST equivalence tests, bootstrap confidence intervals, Bayes factors, the R_E difficulty-weighting robustness check, and the rank-based accuracy cross-check) were computed directly on the released dataset (`anonymized_df.json`) using Python 3.14.5, with pandas 3.0.3, numpy 2.4.6, scipy 1.18.0, statsmodels 0.14.6, pingouin 0.6.1 (Bayes factors), and scikit-posthocs 0.14.0 (Dunn’s post-hoc test). This is a separate environment from the LLM-based SFF scoring pipeline, whose software details are reported in Appx.B. (Table 4).

Age-Group Effects by Color Category Appx. B.2.1 reports that, pooled across all four color categories, Age Group significantly predicted error rate in both the KR and Other cohorts ($p < 0.001$ for both), but not response duration (KR $p = 0.126$; Other $p = 0.592$). Table 5 breaks this down by color category via Kruskal-Wallis H -tests. Error rate differs significantly by age group in every category and cohort (all $p < 10^{-5}$). Response duration shows a less consistent pattern: significant age effects emerge for Black and White in at least one cohort, but not for Red or Yellow. Because splitting the data this finely substantially reduces the per-cell sample size — particularly for the Other cohort, where individual age-by-category groups range from single to low-hundreds of trials, versus consistently larger KR cells — this category-level duration pattern should be treated as exploratory rather than a firm contradiction of the pooled null result above.

Category	Group	H_{err}	p_{err}	H_{dur}	p_{dur}
Black	KR	179.54	1.11×10^{-38}	50.76	5.50×10^{-11}
	Other	27.11	5.59×10^{-6}	3.35	0.341
Red	KR	85.85	1.71×10^{-18}	3.12	0.373
	Other	26.39	7.91×10^{-6}	1.28	0.733
White	KR	273.46	5.51×10^{-59}	22.92	4.19×10^{-5}
	Other	28.99	2.25×10^{-6}	14.85	1.95×10^{-3}
Yellow	KR	195.17	4.66×10^{-42}	2.06	0.559
	Other	34.14	1.85×10^{-7}	2.83	0.419

Table 5: Kruskal-Wallis H -tests for error rate and response duration across age groups, by color category and country group.

Predictor	Coeff	Std Err	z	$P(> z)$	[0.025	0.975]
<i>Intercept</i>	−2.0486	0.179	−11.433	0.000	−2.400	−1.697
<i>Age Effects (vs. Baseline: 18-24):</i>						
C(age_group)[T.25–34]	0.1033	0.054	1.905	0.057	−0.003	0.210
C(age_group)[T.35–44]	0.1710	0.078	2.202	0.028	0.019	0.323
C(age_group)[T.45+]	0.8978	0.055	16.365	0.000	0.790	1.005
<i>Country Effects (vs. Baseline: AU):</i>						
C(country)[T.KR]	0.0056	0.167	0.034	0.973	−0.323	0.334
C(country)[T.UK]	0.1543	0.174	0.888	0.375	−0.186	0.495
C(country)[T.US]	0.1971	0.187	1.054	0.292	−0.170	0.564
<i>Country Effects (vs. Baseline: KR) — pure reparameterization of the identical fit:</i>						
C(country)[T.AU]	−0.0056	0.167	−0.034	0.973	−0.334	0.323
C(country)[T.UK]	0.1486	0.068	2.201	0.028	0.016	0.281
C(country)[T.US]	0.1915	0.093	2.068	0.039	0.010	0.373
<i>Control Variables:</i>						
C(media_type)[T.Video]	0.2172	0.041	5.279	0.000	0.137	0.298
media_hours_per_week	0.0004	0.001	0.299	0.765	−0.002	0.003

Table 6: Multivariate GLM Regression Results for Error Rate (Filtered Data). Model includes the four-level *country* term only; see Appx. B.2.1 for why *country_group* is excluded (exact collinearity with *country*). The two Country Effects blocks are the identical fit reparameterized under two different baselines (AU vs. KR), shown together since each makes a different pairwise contrast directly readable; Intercept, Age Effects, and Control Variables coefficients are unchanged between the two parameterizations.

R_E Difficulty-Weighting Robustness Check The difficulty-level weights $L_d \in \{1, 2, 3\}$ in Eq. 1 are a monotonically increasing proxy for the actual difficulty ordering of the task (Level 1 < Level 2 < Level 3), rather than a value derived from the data. To confirm that the Korean-cohort accuracy advantage is not an artifact of this specific weighting, we recomputed R_E under three alternative monotonically increasing weight schemes and reran the primary comparisons (Table 8). The KR-vs-Other Mann-Whitney comparison remains highly significant ($p < 10^{-32}$) and the GLM country effect remains significant ($p < 0.05$) under every scheme tested, and the direction (lower median R_E for KR) is unchanged throughout.

The same robustness holds for the country-specific pattern underlying the Australia Divergence (§4.1.3, §4.2): under all four weighting schemes, KR remains significantly different from UK and US (GLM, KR-referenced, $p < 0.05$ throughout) but not from AU ($p = 0.86$ – 0.97 across all four schemes), matching the pattern reported for the original $\{1, 2, 3\}$ weighting in Table 6.

Rank-Based Accuracy Cross-Check A natural alternative to R_E ’s linear difficulty weighting is a Discounted Cumulative Gain (DCG)-style ranking metric. However, DCG is designed for ranked retrieval with graded relevance at fixed list positions and does not map onto a block-counting estimation task without introducing additional design choices (e.g., how difficulty level should determine “position,” or what

Predictor	Coeff	Std Err	z	$P(> z)$	[0.025	0.975]
<i>Intercept</i>	2.7206	0.076	36.024	0.000	2.573	2.869
Age Effects (vs. Baseline: 18-24):						
C(age_group)[T.25–34]	−0.0040	0.021	−0.189	0.850	−0.046	0.038
C(age_group)[T.35–44]	0.0311	0.031	1.012	0.311	−0.029	0.091
C(age_group)[T.45+]	0.0441	0.022	2.005	0.045	0.001	0.087
Country Effects (vs. Baseline: AU):						
C(country)[T.KR]	−0.1271	0.066	−1.919	0.055	−0.257	0.003
C(country)[T.UK]	−0.1266	0.069	−1.841	0.066	−0.261	0.008
C(country)[T.US]	−0.1338	0.074	−1.808	0.071	−0.279	0.011
Task-related Variables:						
actual_blocks	0.0264	0.001	20.263	0.000	0.024	0.029
error_rate	−0.0050	0.024	−0.210	0.834	−0.051	0.041
Control Variables:						
C(media_type)[T.Video]	−0.0153	0.016	−0.936	0.349	−0.047	0.017
media_hours_per_week	0.0015	0.001	2.577	0.010	0.000	0.003

Table 7: Multivariate GLM Regression Results for Response Duration (Filtered Data). Model includes the four-level *country* term only; see Appx. B.2.1 for why *country_group* is excluded (exact collinearity with *country*). Note that, unlike for error rate, no country contrast reaches significance in this model (§4.1.3, §4.2).

Weights $\{L_1, L_2, L_3\}$	KR med. R_E	Other med. R_E	MWU p	GLM β (Other)	GLM p
$\{1, 2, 3\}$ (main paper)	0.056	0.147	2.6×10^{-40}	0.152	0.0066
$\{1, 2, 4\}$	0.046	0.135	8.9×10^{-39}	0.150	0.0101
$\{0.5, 1, 5\}$	0.053	0.155	2.5×10^{-33}	0.149	0.0215
$\{1, 3, 5\}$	0.036	0.111	4.0×10^{-38}	0.129	0.0444

Table 8: Robustness of the KR-vs-Other (English-speaking) accuracy comparison to the difficulty-level weighting scheme used in R_E (Eq. 1). The first row is the scheme used throughout this paper. GLM β/p are for $C(\text{country_group})[\text{T.Other}]$ in the age-weighted GLM (§3.5). All schemes are monotonically increasing in difficulty level; none change the direction or significance of the core accuracy finding.

constitutes “relevance”). We instead implement the most direct rank-based analogue, which requires no difficulty-weighting scheme at all: within each difficulty level, we rank all trials by raw proportional error ($|\text{error_amount}|/\text{actual_blocks}$, i.e., without dividing by L_d) and convert ranks to percentiles (1 = most accurate, 0 = least accurate).

Under this fully rank-based measure, the KR-vs-Other accuracy advantage remains highly significant (Mann-Whitney $p = 4.1 \times 10^{-42}$; GLM $\beta = -0.056$, $p = 1.3 \times 10^{-13}$), and the pairwise country pattern underlying the Australia Divergence is preserved: KR is not significantly different from AU ($p = 0.137$) but is significantly more accurate than UK ($p = 7.8 \times 10^{-8}$) and US ($p = 1.9 \times 10^{-9}$)—if anything, more strongly than under R_E itself. This rank-based measure correlates almost perfectly with R_E (Spearman $\rho = 0.971$), indicating that R_E ’s specific weighting scheme is not distorting the relative accuracy ordering driving our conclusions.

Distributional-Family Robustness Check (Gamma vs. Tweedie) The Gamma family used for both GLMs (Table 6; Table 7, Appx. C) formally assumes strictly positive support, but R_E is exactly zero on trials where the participant’s estimate exactly matches the true block count (34.3% of trials)—a point mass the Gamma distribution cannot represent. To verify the reported language effect does not depend on this technical mismatch, we refit the error-rate model under a Tweedie family (power parameter $p \in \{1.3, 1.5, 1.7\}$, log link), which permits a point mass at zero and is the formally appropriate alternative for this data. The language effect ($C(\text{country_group})[\text{T.Other}]$) remains positive and significant across all three Tweedie specifications ($\beta = 0.132\text{--}0.146$, $p = 0.007\text{--}0.012$), closely matching the original Gamma estimate ($\beta = 0.152$, $p = 0.0066$); sign, magnitude, and significance are all stable under the change of

Comparison	Metric	n_a	n_b	Cohen’s d	Rank-biserial r	MWU p
KR vs. Other	error rate	8028	1506	0.174	0.211	2.4×10^{-40}
KR vs. Other	duration	8028	1506	0.101	0.086	1.3×10^{-7}
KR vs. AU	error rate	8028	132	-0.063	0.011	0.826
KR vs. AU	duration	8028	132	0.227	0.196	1.1×10^{-4}
KR vs. UK	error rate	8028	940	0.200	0.222	5.7×10^{-30}
KR vs. UK	duration	8028	940	0.102	0.079	6.7×10^{-5}
KR vs. US	error rate	8028	434	0.181	0.250	2.6×10^{-19}
KR vs. US	duration	8028	434	0.062	0.065	0.022
Text vs. Video	error rate	3472	6062	0.067	-0.030	0.012
Text vs. Video	duration	3472	6062	-0.036	-0.053	1.7×10^{-5}

Table 9: Standardized effect sizes for the major group comparisons reported in this paper. Cohen’s d is computed on raw (unweighted) group values; rank-biserial r is derived from the Mann-Whitney U statistic and does not assume normality, serving as the nonparametric companion given R_E ’s documented non-normal distribution (Appx. B.1). By conventional benchmarks ($|d| \sim 0.2$ small, ~ 0.5 medium, ~ 0.8 large), all effects here are small despite very low p -values in several rows — expected given n in the thousands, and consistent with this paper’s exploratory framing.

Comparison	Margin	TOST p	BF ₁₀	Conclusion
KR vs. AU, error rate	± 0.068	0.003	0.22	Equivalent
KR vs. AU, duration	± 3.345	0.625	2.96	Not equivalent

Table 10: Two One-Sided Test (TOST) equivalence results for the “Australia Divergence” (KR vs. AU), with an equivalence margin of ± 0.2 pooled-SD (Cohen’s $d = 0.2$, the conventional small-effect threshold) converted to raw units for each metric. This margin is a defensible convention, not derived from the data. The error-rate equivalence conclusion is corroborated by a nonparametric bootstrap check: the 90% CI of the median difference, $[-0.032, 0.025]$, falls entirely within the same margin. We additionally report a JZS Bayes factor (Welch two-sample t -test, default Cauchy prior scale $r = 0.707$) so that the null result is supported by Bayesian as well as frequentist evidence: $\text{BF}_{10} = 0.22$ for error rate is moderate evidence *for* the null (equivalently $\text{BF}_{01} \approx 4.6$), converging with both TOST and the bootstrap CI; $\text{BF}_{10} = 2.96$ for duration is weak/anecdotal evidence *against* the null, consistent with TOST’s “not equivalent” conclusion for that metric. Verbal labels follow the conventional Jeffreys/Lee-Wagenmakers scale.

distributional family. Likelihood-based fit statistics (log-likelihood, pseudo- R^2) are undefined under the Gamma family in the presence of exact zeros and are therefore not reported for either family; coefficient estimates, obtained via iteratively reweighted least squares, remain valid regardless and are additionally cross-validated throughout this paper against non-parametric tests that make no distributional assumption at all. We recommend a Tweedie (or zero-inflated) specification as the more formally appropriate choice for future replications of this measure.

Grue-Only GLM Specification The main-text error-rate GLM (Table 6) is estimated on the full, category-pooled dataset, with color category entered as a covariate; this is the better-powered test of the language effect, since it retains all 9,534 trials rather than a single-category subset. As a robustness check, we refit the identical model specification restricted to Grue-category trials only ($n = 293$ English-speaking trials, $n = 1,682$ Korean). The language effect remains directionally consistent (English-speaking cohort less accurate than Korean) but does not independently reach significance at this reduced sample size ($\beta = +0.218$, $p = .085$). This is consistent with the non-parametric Mann-Whitney comparison on the same Grue subset, which is significant ($p < 0.001$): the two tests differ here primarily in statistical power, as the GLM’s covariate adjustment further reduces its effective degrees of freedom relative to the unadjusted rank-based test. We therefore treat the category-pooled GLM, not this Grue-only refit, as the primary covariate-adjusted evidence for the language effect on accuracy, and read the Grue-only estimate as directionally corroborating rather than independently decisive.

SFF Temperature-Robustness Check To address whether SFF scores are stable under changes to the LLM’s generation temperature, we scored a stratified sample of 80 lexical items (preserving the released

dataset’s KR:Other keyword-count ratio) with Gemini 2.5 Flash using the exact prompt in Appx. B.2.4, varying only the temperature parameter. We first established a seed-noise baseline by independently re-running the scoring five times at $T = 0.9$ (the temperature used to build the released dataset); we then compared this against single runs at $T \in \{0.0, 0.3, 0.7\}$, computing the cosine similarity of the resulting (Sentiment, Formality, Friendliness) vectors in each case.

Median cosine similarity was 0.986 across the $T = 0.9$ seed-noise baseline and 0.990–0.991 across all three temperature-variation conditions – comparable to, if not marginally higher than, the baseline (Fig. 7). With three temperature conditions tested, we apply a Bonferroni-corrected significance threshold ($\alpha = 0.05/3 \approx 0.0167$); no condition’s Mann-Whitney comparison against the baseline reaches this threshold ($p = 0.049$ – 0.055 , uncorrected), indicating no statistically significant difference between temperature-induced variation and ordinary re-sampling noise at fixed temperature.

We note that the *mean* similarity (0.926 baseline) is noticeably lower than the median, driven by a small number of low-magnitude (L_2 norm < 0.3) SFF vectors for emotionally weak or neutral keywords (e.g., “busy,” “reference,” “thirsty”), for which cosine similarity is known to be poorly conditioned: trivial absolute score differences between runs can flip the cosine angle substantially even though the underlying scores are essentially unchanged. Restricting to keywords with clearer affective signal (L_2 norm ≥ 0.3) both raises the mean (0.948) and further reduces the already non-significant temperature effect. We report median as the primary statistic, consistent with our treatment of R_E elsewhere in this paper given its similarly non-normal distribution (Appx. B.1.1).

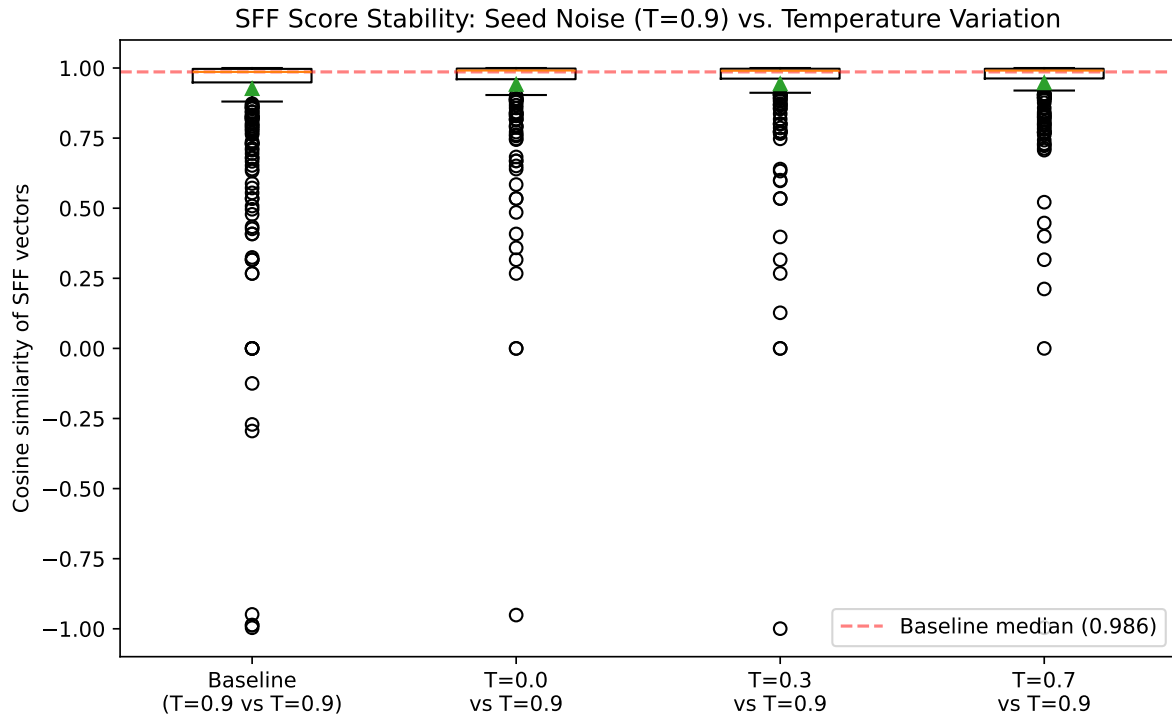


Figure 7: SFF vector cosine similarity: seed-noise baseline ($T = 0.9$ vs. $T = 0.9$, five independent re-runs) compared against single runs at $T \in \{0.0, 0.3, 0.7\}$. Median similarity (≈ 0.99) is comparable across all conditions; low outliers are concentrated in near-neutral keywords (see text).

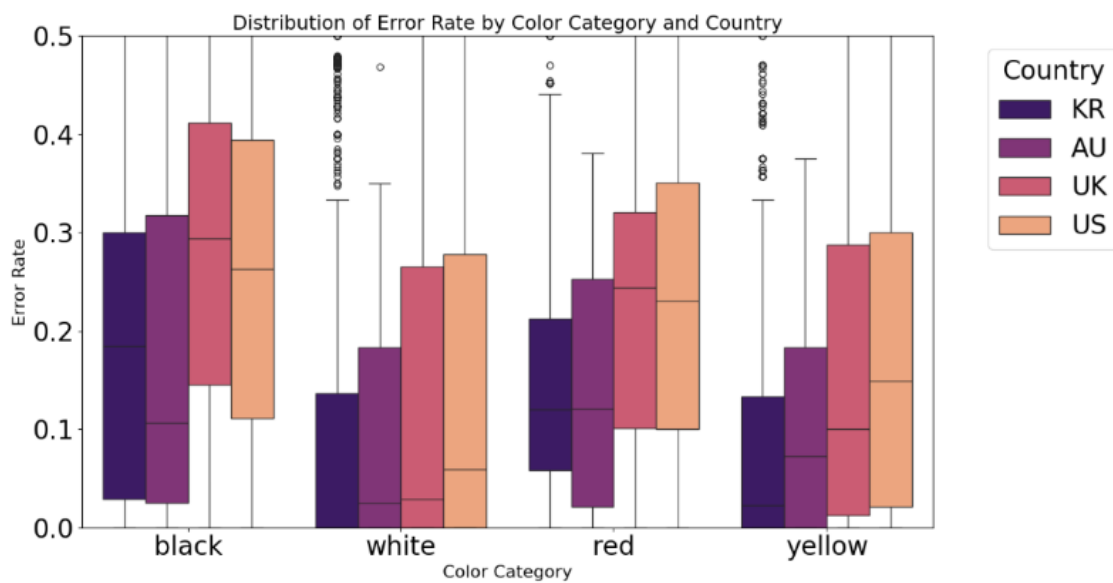
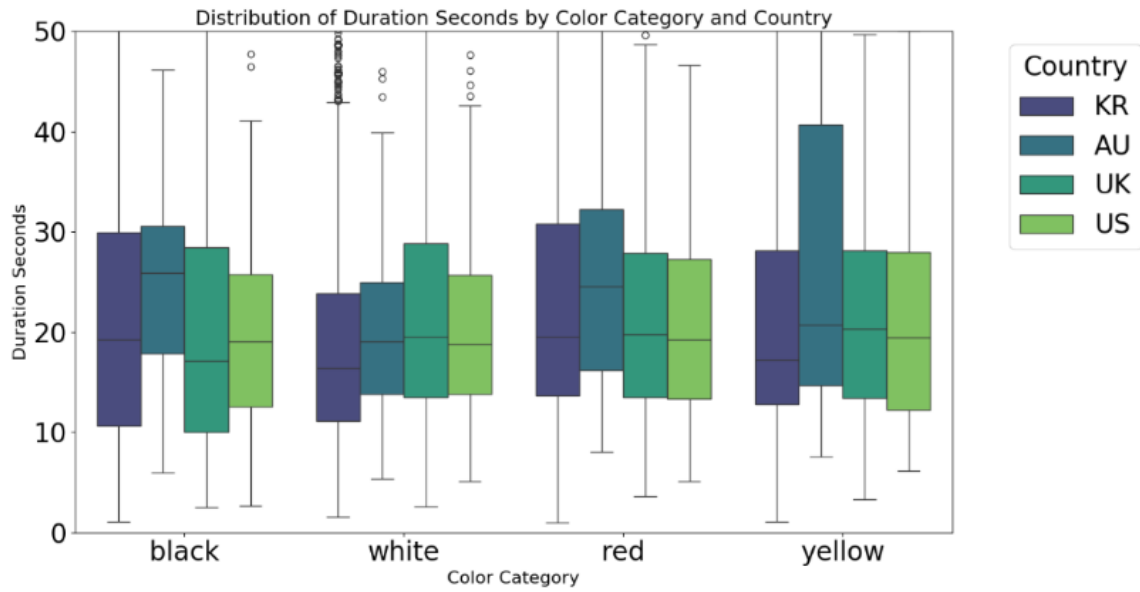


Figure 8: Distribution of Response Duration (top) and Error Rate (bottom) by Color Category and Country, highlighting the performance gap between linguistic cohorts.

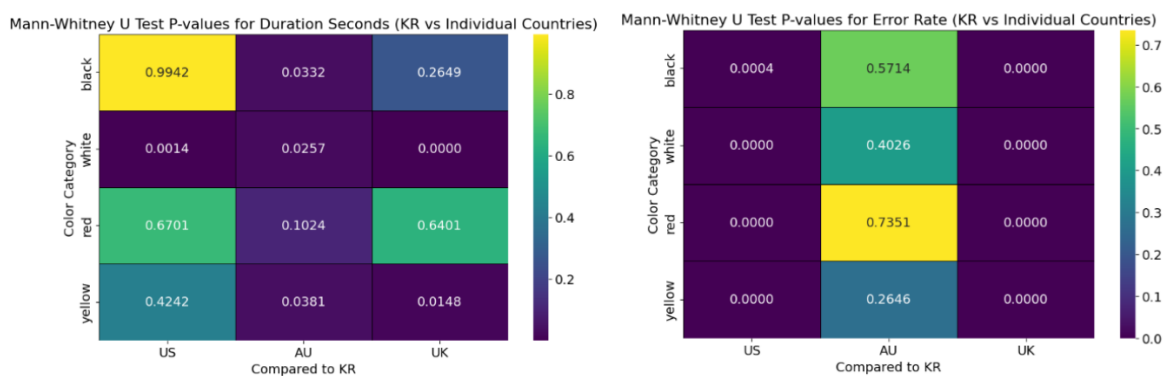


Figure 9: Error rates and durations based on the P value obtained from the Mann-Whitney U test by Countries.

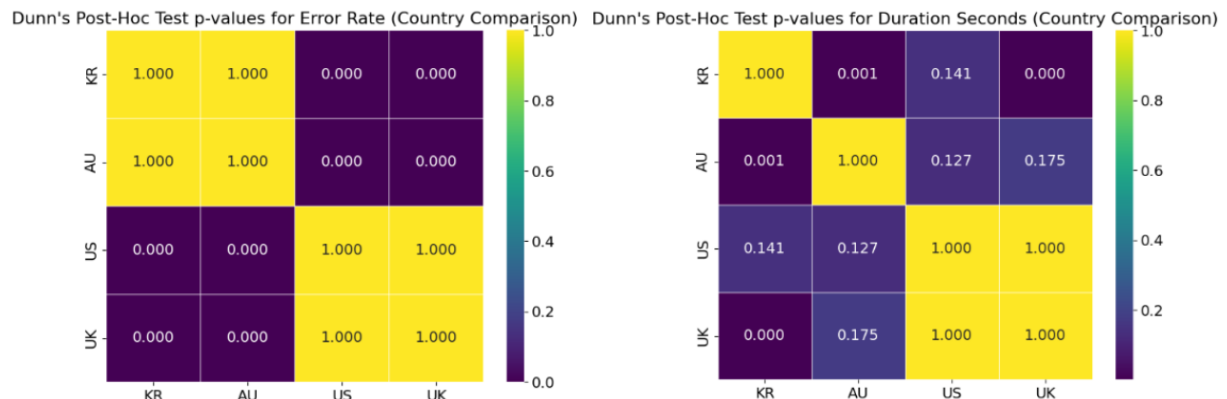


Figure 10: Dunn's Post Hoc Test for Error Rate and Duration Seconds for Cognitive Perception Test (Study 1, Country Comparison)

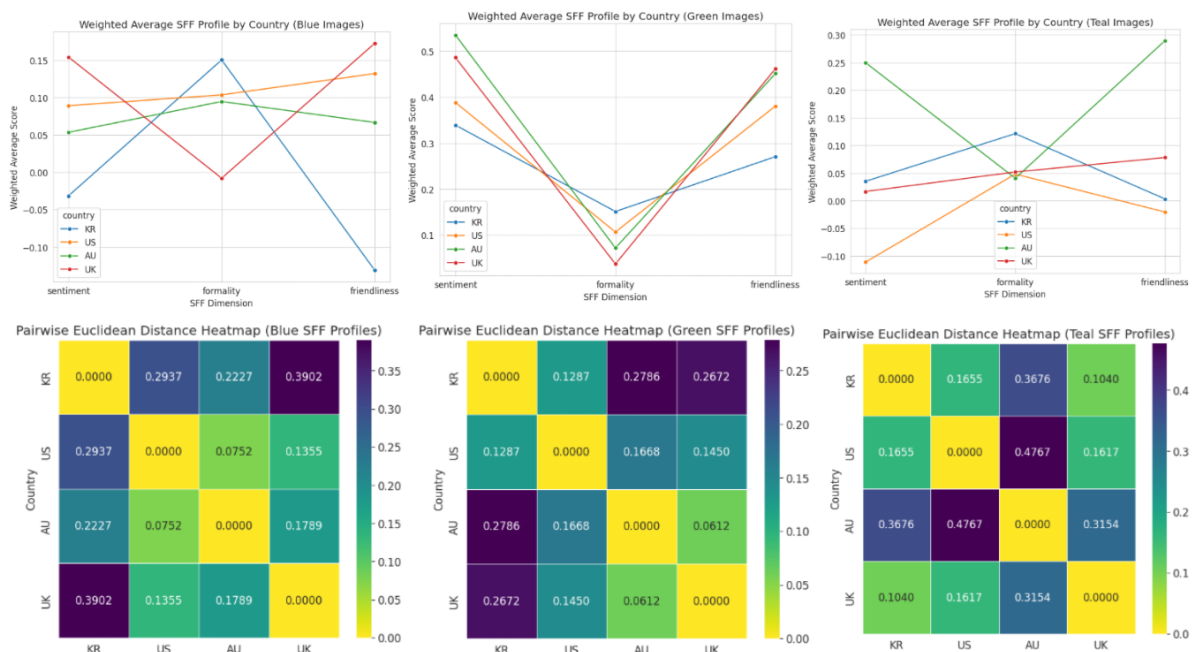


Figure 11: Weighted Average SFF Profile by Country and Pairwise Euclidean Distance Heatmap (Study 2)

Appendix D: Qualitative Case Study: Semantic Network Segregation

To provide a high-resolution view of the affective divergence measured by the SFF framework, we performed a semantic network analysis on a representative stimulus: the 'underwater diver' image (Blue spectrum, Cluster 2).

A note on what follows. This appendix contains two analyses of a different nature that should not be conflated. D.2–D.3 present a *qualitative* semantic-network and word-cloud analysis of keyword co-occurrence for a single representative stimulus. D.1 and D.4, by contrast, concern the *quantitative* K-means clustering of participants' SFF vectors (Fig. 17). As D.1 shows below, this quantitative clustering does not recover participants' country membership at any tested K — the qualitative keyword differences illustrated in D.2–D.3 for one stimulus should accordingly be read as an illustrative case study of lexical variation, not as evidence that the quantitative clusters themselves are organized by country.

D.1 Cluster Validation: Elbow and Silhouette Analysis

To motivate the choice of $K = 3$ for the K-Means affective clustering, we computed both the within-cluster inertia (Elbow Method) and the Silhouette score across $K = 2$ to $K = 10$ on the standardized per-participant, per-color SFF vectors (Fig. 12). The inertia curve decreases smoothly without a sharp bend, and the Silhouette score is in fact marginally higher at $K = 2$ than at $K = 3$. We nonetheless report $K = 3$ throughout, as it offers a more interpretable three-way archetype structure (see Fig. 17) and corresponds to a visible, if gradual, elbow; we flag $K = 2$ as a plausible alternative for future robustness checks.

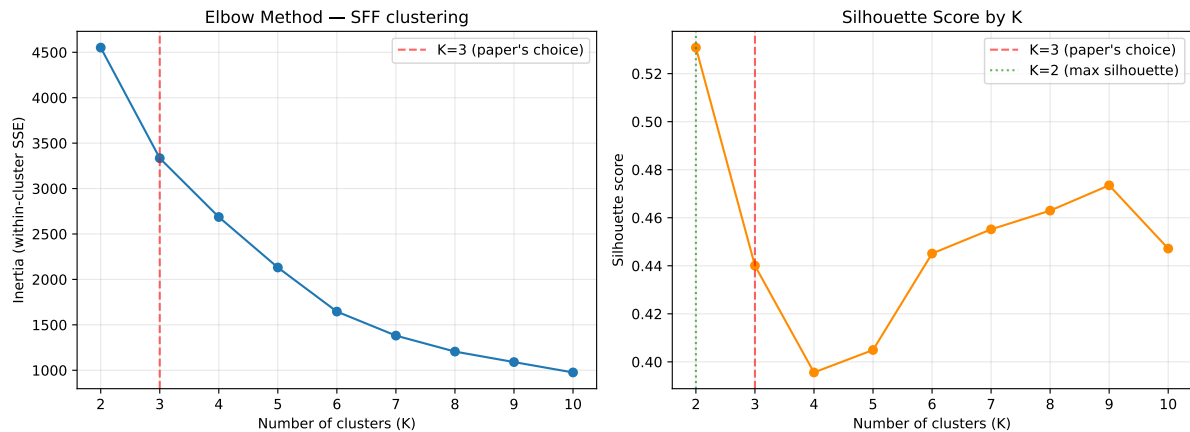


Figure 12: Elbow Method (inertia) and Silhouette Score for K-Means clustering of per-participant, per-color SFF vectors ($K = 2$ to 10). $K = 3$, used throughout this paper, is marked in red; $K = 2$, the Silhouette-optimal choice, is marked in green.

Does the clustering recover country membership, at any K ? We additionally computed the Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI) between the K-means cluster assignment and participants' country label, for every K from 2 to 8 (Fig. 13), with no post-hoc selection of K . Agreement is negligible throughout ($\text{ARI} \leq 0.006$, $\text{AMI} \leq 0.017$, on a scale where even weak agreement is conventionally ~ 0.1 – 0.2), and no K stands out. A permutation-null check (country labels shuffled 500 times, preserving class sizes) confirms these values are trivial in absolute magnitude regardless of where they fall relative to their own null distributions. Plain, uncorrected Normalized Mutual Information (NMI) is not shown here because it exhibits a known small-sample upward bias — it appears to rise toward $K = 6$, but AMI, which is specifically constructed to correct for this bias (analogous to how ARI corrects the plain Rand Index), does not, confirming the rise is an artifact rather than signal. **This is a K -independent finding, not an artifact of the specific $K = 3$ used elsewhere in this paper:** the three-axis SFF space does not appear to contain enough information to separate participants by country via clustering, at any cluster count tested. This finding is discussed further in the Limitations section ("Feature-Space Sufficiency for Cultural Clustering") and directly informs how the cluster labels in D.4 should be read.

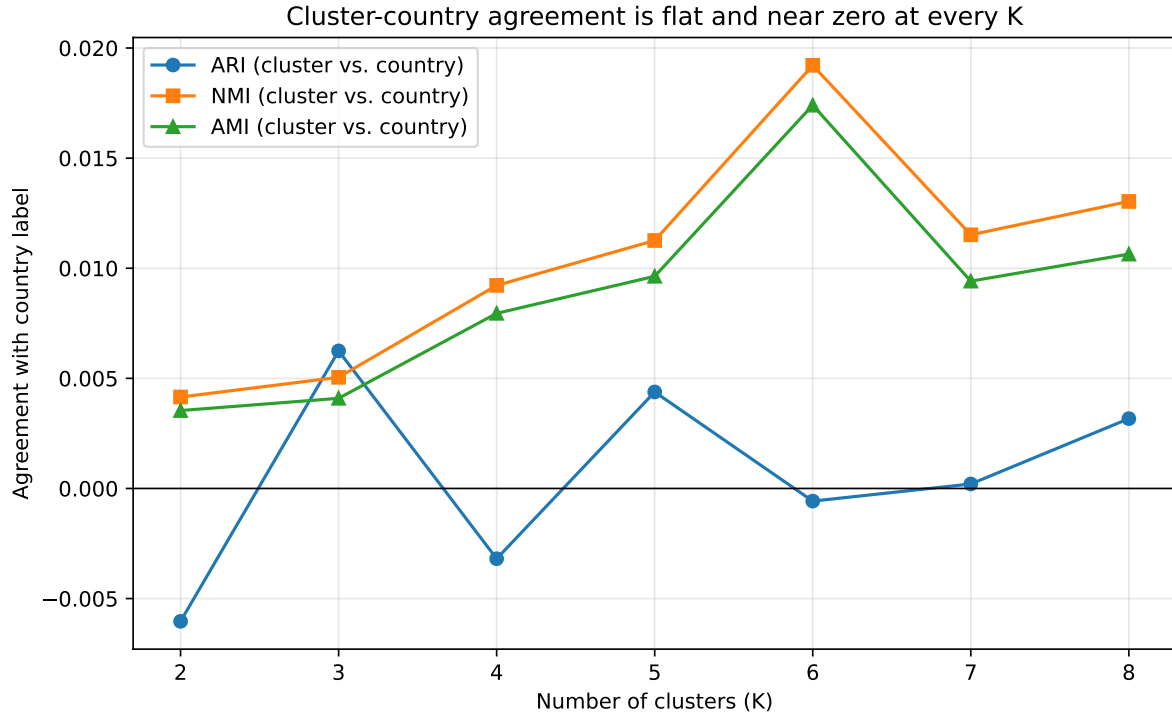


Figure 13: Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI) between K-means cluster assignment and country label, $K = 2$ to 8. Both remain near zero throughout; no K recovers country structure.

D.2 Structural Isolation of Linguistic Networks

As illustrated in Fig. 14, the keyword networks of the two linguistic groups exhibit total structural segregation for this representative stimulus. This isolation suggests that the underlying keyword-level affective responses collected for this specific image are not merely variations of a shared emotional map, but represent two distinct sets of lexical associations — a stimulus-level, qualitative observation distinct from the quantitative clustering result above (D.1).

- **Korean Network (Right):** Forms a dense, highly integrated cluster. It is centered on abstract and sophisticated descriptors such as '*Shinbirowoon*' (mysterious), '*Chagawoon*' (cold/chilly), and '*Siwonhan*' (cool/refreshing).
- **English Network (Left):** Forms a significantly smaller, isolated cluster relying on basic sensory and emotional terms such as '*happy*', '*peaceful*', '*tranquil*', and '*cold*'.

D.3 Lexical Density in Cluster 2

The word cloud in Fig. 15 further details the lexical composition of this segregated space. The prominence of terms like '*Shinbirowoon*' (mysterious) and '*Siwonhan*' (cool-positive) suggests that for the Korean cohort, the blue spectrum evokes complex, multi-layered social emotions that transcend the simpler affective mappings observed in the English cohort.

D.4 Qualitative Clusters for Teal/Mint Spectrum

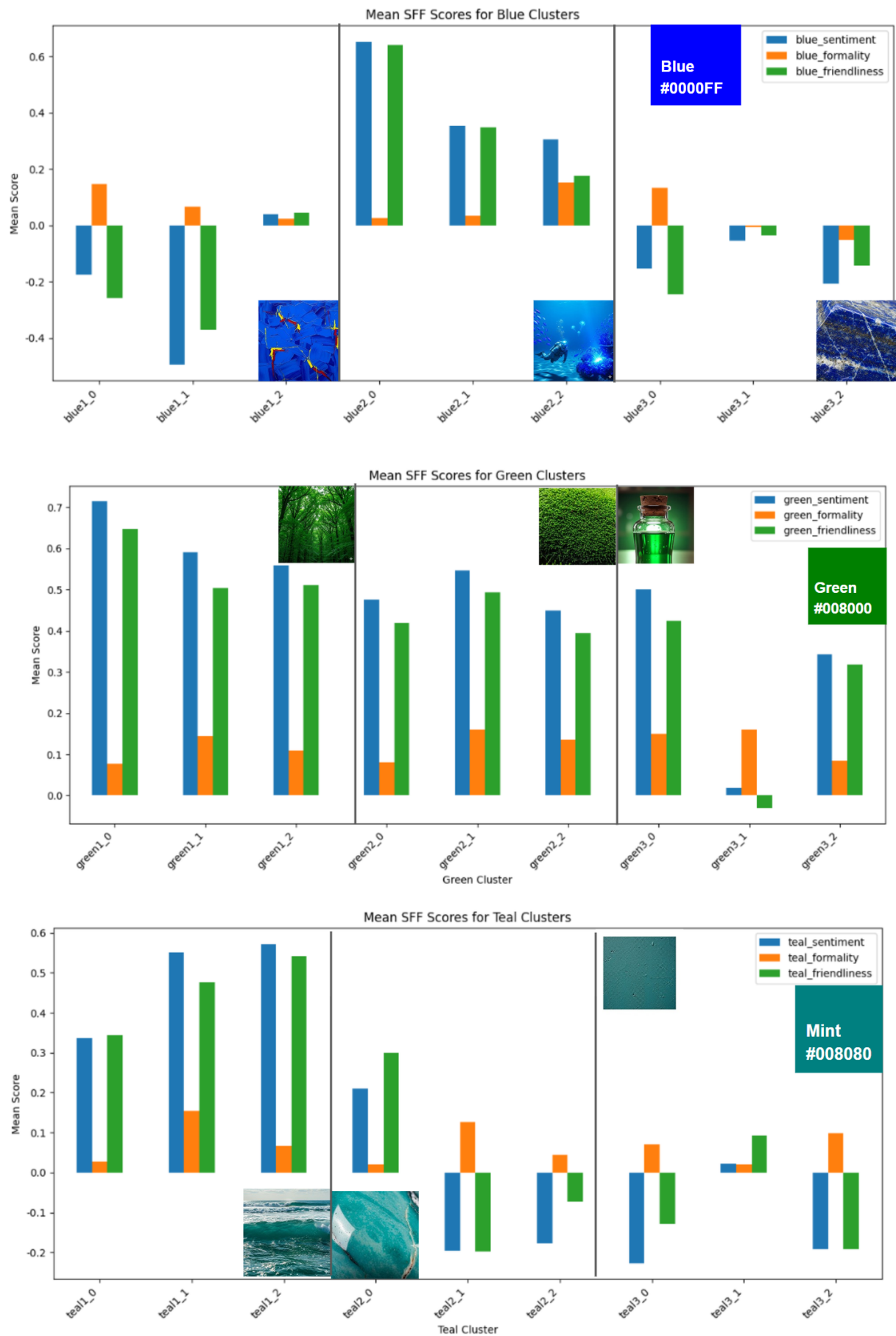


Figure 17: Mean SFF Scores by Color Clusters